When the Model Screens Off the Harness: Per-Decision Cross-Layer Causal Credit in LLM Agents

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Abstract

An LLM agent is a model wrapped in a harness: the program that manages its context, retries its failures, and decides when it stops. The two layers decide jointly, but outcome-level RL and single-layer credit methods cannot tell their contributions apart. We build the instrument that can: deterministic counterfactual replay with a noise floor measured at exactly zero, re-executing a recorded trajectory with exactly one decision changed and reading off per-decision credit for the harness, the model, and their interaction, on the same trajectory. The instrument reveals a phenomenon we call screening-off: given budget slack, the model's next action absorbs the harness decision's damage at 53 of 56 reward-pivotal points — 24 of 31 under an information-poor alternative that cannot re-inject what the harness destroyed, so the two estimands bracket the phenomenon rather than settling it — so single-layer harness credit — the estimator implicit in prior work — systematically bills the harness for effects that belong to the model. Whether screening holds is triple-dependent (model, harness, task): breaks recur at fixed points across configurations; a stronger model breaks outright. Screening has a training consequence we state as a no-free-lunch proposition: under matched compute, screened points contribute exactly zero billed per-decision signal at strictly positive replay cost — so a cheap pre-training census decides whether credit can pay for itself before any training is run. Four pre-registered training rounds confirm the closed branch of that rule the hard way: outcome-only RL beats replay credit at matched dose. On a second, richer agent class (multi-file tasks, five interacting harness decision types), screening breaks and the census licenses training under three of four registered accountings — and the licensed training collapses onto a fixed attractor before credit ordering becomes testable. Replay credit is worth its cost for measurement; whether it is worth training on is a measurable property of the agent, not an article of faith. We release the apparatus and a 34-item registration ledger whose evidential force rests on its reported failures.

1 Introduction

An LLM agent is not a model. It is a model wrapped in a harness: a program that manages context, retries failed calls, and terminates episodes. The harness makes as many decisions per episode as the model does, and those decisions — especially context management — destroy or preserve the information the model conditions on. Yet outcome-level RL assigns one scalar return to every decision regardless of layer, and per-step credit methods attribute within a single layer only. If a harness decision was harmless because the model's next action compensated for it, single-layer credit bills the harness for an effect the model screened off — a double-counting error invisible unless both layers are measured on the same trajectory.

We ask the question directly: on the same trajectory, what is the causal contribution of a harness decision, of a model decision, and of their interaction? We answer with deterministic counterfactual replay: re-execute a recorded trajectory from any decision point with exactly one decision changed and the prefix forced, defining per-decision credit $C(d) = R_{\text{orig}} - R_{\text{cf}}$;

pair and joint flips yield C_H, C_M, C_{HM} and the interaction $I = C_{HM} - C_H - C_M$. Because replay is exactly deterministic (measured, not assumed — and silently broken by vLLM's automatic prefix caching), every nonzero credit is signal.

The headline finding: whether interaction vanishes is a property of the (model, harness, task) triple. On our primary triple it is regime-dependent: with budget slack the model action screens off the harness decision exactly — $C_{HM} = C_M$ at every measured point (57/57 raw; 53/56 under the stricter reward-pivotal endpoint; §4.3 maps every count to its endpoint); the breaks recur at fixed decision points across configurations (under pressure, 8/65 points — though only 3 tasks break, so clustered inference is weak), co-occurring with reward saturation: interaction is detectable only inside a competence window where the agent neither aces nor fails everything. A stronger model breaks at a flat rate in both regimes (§8).

We then test whether interaction-aware credit improves training. Screening has a budget consequence (Proposition 2): under dose-matched compute, screened points contribute exactly zero billed per-decision signal at strictly positive replay cost, so whether credit can pay for itself is a gate measurable before any training by the same census that established screening (Corollary 3). Four training acts validate this the hard way: three identified failures, then, under a repaired objective, the first arm separation — outcome-only wins at matched dose; corrected credit, correctly zero where the model shields, loses the gradient the single-layer bias was providing.

Contributions.

1. The instrument. Deterministic counterfactual replay with a noise floor measured at exactly zero under explicit serving premises, yielding C_H, C_M and I per decision on the same trajectory — plus a cautionary finding: vLLM's automatic prefix caching silently breaks greedy-decoding determinism.
2. The finding. Screening-off, with its anatomy: a double-counting identity ($C_H = -I$ at screened points; $C_{HM} - C_M$ is the correction), two verified mechanisms, and a replication record scoping the shield to the (model, harness, task) triple.
3. The rule. A no-free-lunch proposition and census gate (Proposition 2, Corollary 3), both branches exercised through training: closed on the single-file stack (four rounds; outcome-only wins at matched dose), and on the five-decision multi-file stack the gate opens under three of four registered accountings, the licensed training runs, and the optimization collapses onto a fixed attractor before credit ordering becomes testable (§7).
4. The record. A 34-item pre-registration ledger with failures reported as results, control-task construct validation, and a claims table (Table 4, Appendix A) in which every number is auditable against the ledger (Table 7).

This is a measurement-and-falsification paper, not a method paper.

2 Related Work

Counterfactual replay for agent credit is an active area, but every existing system we know of attributes within a single layer. CHILL-Harness (Fu et al., 2026) intervenes on harness decisions, training an amortized effect estimator from checkpointed paired replay, but never measures the model layer or the interaction; our single-layer arm (§6) reproduces its principle, turning the comparison into a result. CAR (Shah, 2026) formalizes per-step do-operations with Shapley attribution under a replay budget — including step–step interaction — but within one decision stream. Composing the two would not yield our measurement: I requires the joint arm C_{HM} — a paired flip of both layers at one point — which no union of single-layer tools defines; the identities we test ($C_H = -I$; additivity under a'_s) are statements about that arm. C3 (Chen et al., 2026a) audits credit fidelity against exact replay ground truth (the validation metric we adopt) but decomposes across agents, not across the model/harness boundary inside one agent. A recent negative result (Zhang, 2026) shows popular step-level credit signals fail to beat chance against replay ground truth, setting the evidential bar our critic section inherits. Positive turn-level results — TCPO (Liao et al., 2026) among them — train dense per-turn signals without replay ground truth or a harness layer: our negative result prices exactly the regime — screening — their settings never measure. DFAH (Khatchadourian, 2026) measures trajectory determinism as an audit property and finds it generally unattained; we engineer the zero floor and build causal credit on it. Co-Harness (Chen et al., 2026b), HarnessCompass (Zhang et al., 2026) and HASE (Luo et al., 2026) jointly optimize harness and weights without a per-decision causal signal. Agent Lightning (Luo et al., 2025) provides rollout infrastructure without attribution. The correction rule $C_{HM} - C_M$ sits in an older lineage: difference rewards and COMA (Wolpert & Tumer, 2001; Foerster et al., 2018) marginalize one agent's action against a baseline; we marginalize one layer's decision against a measured joint counterfactual, made exact by the zero floor. Hindsight credit assignment (Harutyunyan et al., 2019) and process reward models (Lightman et al., 2024) assign per-step credit as learned, amortized quantities inside the model layer — where our transfer test shows amortization failing. Table 6 (Appendix G) summarizes the gap: no prior work measures C_H, C_M and I at the same decision point against an exact zero-floor replay.

3 Setup: A Two-Layer Agent with Deterministic Replay

3.1 Agent, decision schema and recorder

The agent is deliberately minimal: a coding agent in a sandboxed environment that reads a task description, writes files, and runs tests. Two layers make decisions. The model layer (Qwen3-4B served by vLLM 0.8.5.post1 on a single NVIDIA RTX 4090, greedy decoding) chooses tool calls: write_file, run_tests, read_file, finish. The harness layer is a scripted policy that at every turn makes a context_policy decision (keep_context vs. summarize_context, the latter compressing history to a fixed-length prefix) and a termination decision. Every decision — model or harness — is recorded with a typed schema: decision id, layer, state before, available actions, chosen action, observation, token costs, and parent/child links. Reward is deterministic: the fraction of hidden pytest tests passed, so $R \in [0, 1]$ with resolution set by the number of tests. Cost-adjusted reward is $R_{\text{eff}} = R - \lambda \cdot \text{prompt tokens}/10^5$. The measurement runs and training Acts 1–3 use $\lambda = 25$; Act 4 and the V2 stack re-derive λ^* analytically (5 and 0.2) from already-measured (R, tokens) pairs, because λ does not transport across token magnitudes (§7). Each value is fixed before the run it governs and never fit to that run's results; the artifact stores raw token totals, so R_{eff} at any λ is recomputable.

3.2 Replay and interventions

Replay re-executes a recorded trajectory through the live agent loop with a forced-action queue (Figure 1): recorded decisions are matched (on type and state hash) and forced up to the intervention point; from there the loop runs free. An intervention replaces exactly one decision (or, for C_{HM}, one harness–model pair). Model-layer alternatives a' are sampled once per point by a fixed procedure (temperature 0.8, seed sequence 2001–2008, first parseable action differing from the recorded one, conditioned on the recorded un-summarized state). Every model-layer credit is thus an estimand relative to this single alternative — $C_M = C(M; a')$ — and sensitivity to re-sampling is measured by a pre-registered robustness study (§4.3). The null intervention — force everything, change nothing — must reproduce R exactly; §4.1 shows it does. All replays run sequentially against a single vLLM instance with a fixed config.

3.3 Tasks

We use 30 synthetic coding tasks in four designed strata plus 10 held-out tasks. V2/L tasks place critical constants beyond the first 240 characters of history, so a summarize decision is informationally destructive at known positions (L are longer variants); S tasks require harness–model cooperation by design; C tasks are controls (all needed information survives

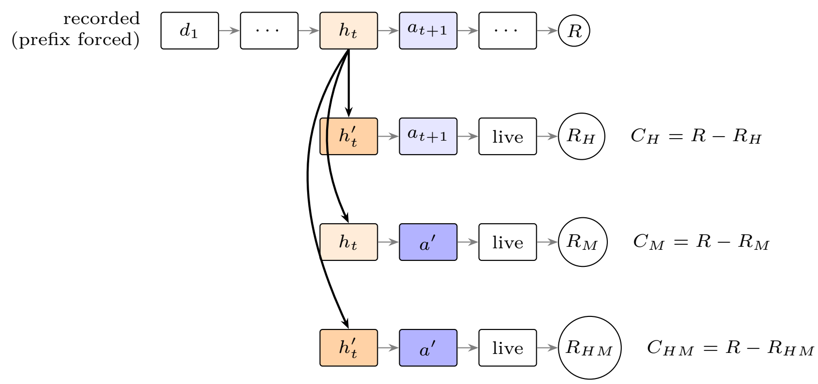

Figure 1: The three-arm replay at one interaction point. The recorded prefix is forced up to the harness decision h_t; each arm changes exactly one thing (the harness decision, the model action, or both) and runs live from there. Interaction is $I = C_{HM} - C_H - C_M$; screening-off is the event $C_{HM} = C_M$ — the joint arm collapses onto the model arm, so the harness flip adds nothing beyond the model's alternative.

any summary, predicting $C_H = 0$). Reward saturation ($R \in \{0, 1\}$) is endogenous to the tests per task; we analyze its consequences explicitly (§4.3, §8).

4 Measuring Cross-Layer Credit

4.1 Noise floor

Every causal claim below is only as good as the replay, so we measure the floor rather than assume it: the pre-registered floor per configuration is $\max |\Delta R|$ over null replays (replay with no intervention; defined on R, not trace equality — 3 nulls took one extra retry while reproducing R exactly). Across 26 dedicated null-replay runs — three models (Qwen3-4B, 1.7B, 8B), three environments (designed pool, MBPP+, HumanEval+), the hardened/curated pools, and an LLM-summarizer cell — plus 1,568 nulls attached to the 24 interaction runs, every one of 9,296 null replays reproduced R exactly: the floor is 0.0 in all configurations (Figure 3a; Table 8 reconciles every count against the run artifacts). This is a measurement under explicit premises — fixed serving config, strictly sequential requests, prefix caching off — not a guarantee.

The premises earn their keep: three incidents, each root-caused and re-run, show the floor is fragile to any unaudited byte or state — an ASLR address leaking through a pytest repr (2/432 nulls broken in a discarded pre-fix run), an accidental concurrent chain ($\Delta R = +0.67$), and a greedy LLM summarizer replacing the rule-based one (floor survives: 317/317 exact). Appendix J reports all three in full. The sharpest premise is prefix caching. vLLM enables automatic prefix caching (APC) by default; with APC on, prefill numerics depend on KV-cache state left by previous requests, so greedy decoding diverges across identical prompts even under strictly sequential requests. In our APC-contaminated run 7/270 nulls were inexact and the floor rose to 0.417 (Figure 3b) — enough to swamp most true credits. Sequential execution is not sufficient; APC must be disabled, and any replay-based attribution built on default vLLM settings inherits this silent failure. Known serving-nondeterminism discussions focus on batching-induced kernel effects (He & Thinking Machines Lab, 2025); APC is distinct — cross-request state — and our contribution is to quantify it as a replay hazard for causal attribution, not to claim its discovery.

4.2 C_H and C_M

With an exact floor, per-decision credits are directly interpretable: any $C \neq 0$ is a real causal effect of that single decision. Figure 4 (Appendix G) shows the distributions over 248 harness points and 66 model points, by stratum. Three regularities matter downstream:

Endpoint (one line)	Count	Note
Raw equality $C_{HM} = C_M$, slack, all measured points	57/57	headline mechanism; g450/g600/g900
Reward-pivotal sign test (pre-reg 10)	53/56	abstract's number; CI95 $[0.87, 1.00]$ task-clustered; endpoint re-specified post hoc
Discovery / confirmation split	14/14 / 39/42	g600 predates registration; strictly post-registration 27/30
a' re-sampling, 3 disjoint seed schedules (pre-reg 21)	99/101	stable outcome met; 79/81 on genuinely new a'; mt6 breaks persist 9/9
Estimand control a'_s, summarized state (pre-reg 26)	24/31	partial: CI95 $[0.69, 0.88]$; all 7 de-screened show re-injection
Pressure regime, raw	8/65 breaks	shield cracks at recurring points

Table 1: Screening scoreboard: every screening-off count in the paper, with its endpoint. "Screened" counts points with $C_{HM} = C_M$; the pressure row counts breaks.

direction is everything (flipping keep→summarize at an information-critical point yields C_H up to $+0.89$ while the reverse is near zero, so we never aggregate across directions); credits are sparse, with mass concentrated at the information-destruction points predicted by task design; and C_M at matched points is frequently nonzero when C_H is not (§4.3).

4.3 Interaction I and its anatomy

The central measurement flips a harness decision and a model decision at the same point, alone and jointly. The model action screens off the harness decision when the joint flip yields exactly the model flip's reward, $C_{HM} = C_M$: conditional on the model counterfactual, the harness flip has no marginal effect (term borrowed from Reichenbach, 1956; Appendix B).

Configuration names: gN = slack (max 12 turns, summarize threshold N tokens); mt$K$ = pressure (max K turns, threshold 600); q8/q17 = Qwen3-8B/1.7B (no prefix = 4B); mbpp/he = external environments; cur/ls = curated pool / LLM summarizer. Screening-off is measured under several pre-registered endpoints, so the scoreboard comes first (Table 1); every count below appears there.

With budget slack (g450/g600/g900) the model action screens off the harness decision exactly: $C_{HM} = C_M$ in 57/57 points — the alternative model action re-establishes what the harness destroyed. Under pressure (max 4/6/8 turns) screening breaks in 8/65 points (slack vs. pressure $p = 0.005$ point-level hypergeometric; $p = 0.125$ task-clustered sign test over 3 breaking tasks; we report both). Breaks recur at the same points across configurations (l_log_parser idx 8 in 3/3 pressure configs; two further points in 2 each) — what a real mechanism, not noise, predicts.

The dependence on pressure intensity is non-monotonic: break rates are 0/21 (mt12), 4/22 (mt8), 3/21 (mt6), 1/22 (mt4) — an inverted U; our pre-registered monotonicity test failed (Cochran–Armitage $p = 0.38$). The strongest correlate is reward saturation: at mt4 the agent fails most tasks outright, so no flip can make things worse and interaction is masked. Conditioning on non-saturated points sharpens the contrast (slack 0/12 vs. pressure 6/19, $p = 0.037$) and partially restores the mt4 rate; these conditioned rates condition on a post-treatment variable and are descriptive, not causal, and their task-clustered versions are weak (§8). We call the picture a competence window: cross-layer interaction is detectable only where the agent neither saturates success nor failure (Table 2; Figure 5, Appendix G).

Anatomy and robustness (summary). Appendix C gives the full record: on the falsifiable subpopulation (reward-non-pivotal points screen by construction, Remark 2) the screened fraction is $53/56 = 0.946$, CI95 $[0.87, 1.00]$ task-clustered, surviving the discovery/confirmation split (confirmation set 39/42) and a' re-sampling under three disjoint seed schedules (99/101 informative slack draws; the three pressure breaks re-break 9/9). Two verified mechanisms — structural re-injection (a', sampled from the un-summarized state,

Model	Pool	Slack	Pressure	Reading
4B	original (30)	0/57	8/65	breaks gated by pressure
1.7B	original (30)	0/23 combined		slack replicates; pressure saturated
8B	original (30)	5/28	6/34, 7/35	flat: regime-independent breaks
4B	curated (22)	4/16	2/23	breaks in both regimes
8B	curated (22)	3/27	3/31	flat again
4B	MBPP+ (60)	0/66 (11/11 pivotal)		screens unconditionally
1.7B	MBPP+ (60)	0/44 (24/24 pivotal)		screens unconditionally
4B	HumanEval+ (60)	0/76 (42/42 pivotal)		screens unconditionally

Table 2: Synthesis: raw screening-off break counts by (model, pool, regime). Every claim about when the shield holds reduces to this grid: pressure-gating is a 4B-on-original-pool phenomenon; the 8B breaks at a flat rate wherever it is in its competence window; MBPP+/HumanEval+ screen everywhere. Counts are breaks/points; detail in Figure 5 (Appendix G).

carries the destroyed information across the cut; 23/24 annotated) and downstream re-summarization (22/52) — explain 72/75 screened points in the recorded traces.

Consequence for credit assignment, formally. At every shielded point single-layer C_H bills the harness for an effect the model absorbs, and the error is exact: the double-counting identity $C_{HM} - C_M = C_H + I$ gives $C_H = -I$ at screened points (Proposition 1) — in the slack regime, all measured points. The coalescence lemma (Lemma 1) licenses the correction rule $C_{HM} - C_M$ at one extra replay per point, and the no-free-lunch bound (Proposition 2) organizes §6 and §7 — all point identities under the determinism premises of §4.1 (Appendix B).

Controls. On control tasks (summaries destroy nothing by construction) the method returns $C_H = 0$ in 11/12 measured points; the exception is a behavioral formatting pathway, not an informational one (Appendix B).

5 The Structure of Harness Credit

Before paying for replays at training time, can C_H be predicted from cheap features, reserving replay for calibration? We trained linear and GBM critics on 248 ground-truth points (features: turn, context size, position, test progress; task-clustered evaluation) against the baselines mandated by (Zhang, 2026): position and context-size heuristics. The answer is negative: the critic never beats the heuristics — pooled Spearman 0.718 vs. 0.770 for context size, and no stratum reverses the order (Appendix H). Harness credit is mostly a function of context size and position — exactly what the information-destruction mechanism predicts — so practitioners get most of the ranking value of C_H from features they already log, reserving replay for calibration and for the interaction term, which no cheap feature predicts. A critic trained on all designed-task C_M ground truth transfers to the 94 MBPP+ points at chance (Spearman 0.03/−0.08; pre-registered secondary of pre-reg 16): amortized credit does not cross environments here.

6 Training with Cross-Layer Credit

Does better credit train a better harness? No — and the reason is the paper's rule at work (Table 5, Appendix F). We train the harness context_policy (a logistic policy over 5 features) with four arms, dose-matched in total rollouts (episodes + credit replays): (1) outcome-only policy gradient; (2) single-layer C_H credit — an explicit reproduction of the CHILL-Harness principle; (3) interaction-corrected credit $C_{HM} - C_M$; (4) a zero-credit control (no replays; under our greedy tie-break, de facto an untrained keep-always policy). The registered criterion (GATE 3) required arm 3 to beat arms 2 and 4. Calibration established the target is learnable: on held-out tasks a context-size threshold policy (thr600) achieves $R_{\text{eff}} = 0.398$, dominating keep-always (0.237) and summarize-always (−0.464), and is policy-expressible.

Acts 1–3 (summary). Appendix F reports Acts 1–3 in full. What survives into Act 4: the estimator is exonerated (pre-registered audit: 58/60 sign agreement on fresh replays, zero harmful flips); the optimization and training distribution are repaired (centering, lower lr, clipping; margin-verified task selection); and Act 3's nine-of-nine collapse to summarize-always identified the objective itself as mis-specified — under $\lambda = 25$ universal cheap failure was the true optimum, so a sane objective must verify the margin against every expressible fixed attractor.

Act 4: sane objective — the arms finally separate, and the bias is load-bearing. The last run (registered with declared outcomes before any rollout) repairs the objective: we measured the third attractor on all 52 tasks, then selected analytically — before any training — the cost weight $\lambda^* = 5$ and the 12 tasks on which thr600 strictly dominates both fixed attractors (held-out window: keep-always 0.392, thr600 0.455, summarize-always -0.031). For the first time, arms 2 and 3 separate: outcome-only reaches held-out $R_{\text{eff}} = 0.440$–0.443, single-layer C_H reaches 0.398–0.405, and interaction-corrected $C_{HM} - C_M$ stays exactly at keep-always (0.392, indistinguishable from the untrained control) — in 3/3 seeds. Our registered secondary hypothesis predicted the opposite ordering (phantom credit should hurt the C_H arm); it failed, and the failure is the finding. The double-counting bias is load-bearing: at screened points the corrected signal is (correctly) zero, so the corrected arm receives almost no gradient precisely where the harness acts, while the biased C_H — whose value there is pure interaction — pushes the policy off the keep-always attractor, in this landscape usefully. At matched LLM-call dose the replay tax is decisive: the credit arms trained on 139/70 episodes against outcome-only's 280, and outcome-only won outright. The flagged confound — sample size, not signal quality — was closed by a registered episode-matched control (pre-reg 31): the outcome-only θ-trajectory stopped at the credit arms' episode counts still reaches 0.398–0.450, $\geq$ the C_H arm in 3/3 seeds (strictly above in 2/3; one seed ties exactly), and beats the corrected arm's attractor in 2/3 seeds at half its episodes. The verdict is identified, not confounded: per-decision replay credit costs more compute than its signal adds, and correcting its bias removes what little training signal the bias was providing. Measurement remains the right use of the replays; billing them to the training loop is not — and the census could have predicted the failure (Corollary 3).

7 The Other Branch of the Rule: A Second Agent Class

If screening-off were an artifact of the single-file agent's simplicity (one dominant harness decision), the rule would have nothing to price elsewhere. We therefore built a second agent class and ran the census as a decision procedure, its gate (Corollary 3) registered up front.

Stack. Mini-SWE tasks: a frozen pool of 60 multi-file repos (15 families $\times$ 4 bug variants; 4–6 source files, 150–300 LOC, 6–12 pytest tests; canonical fix passes 100%, buggy state scores strictly inside $(0, 1)$; determinism validated 3$\times$ at generation). The harness makes five decision types (context policy, observation formatting, test scheduling, retry, termination), each logged and forceable in replay. Two configurations (slack and pressure) give 120 baseline episodes. A pilot finding: under a JSON action envelope the model emitted zero write_file actions in 437 calls; plain text restored normal behavior — the envelope is itself a first-order harness decision. Its estimand-level face: plain-text actions are near-deterministic, and the a' sampler failed at 87/114 points until a pre-registered escalation to temperature 1.2 (strata diverge 0.084 $<$ the declared 0.15; pooled analysis primary).

Floor and pivotality. The zero floor survives the class change (117/117 nulls exact, including three retry-bearing trajectories). Selective screening (693 replays, $\leq$6 points per trajectory, $\leq$2 per type) found 114 reward-pivotal points; all five decision types are pivotal in the slack regime.

The census verdict. At 48 valid census points (of 114 pivotal: 65 excluded for want of an a' even after escalation, 1 for a retry-preceded span) the pre-registered outcome bin is s3, general break: no type screens at ≥ 0.90 — context policy 0.609 ($n = 23$), observation

Non-screened mass	with duals	without duals
over measured points	$24/48 = 0.50$ (opens)	$15/38 = 0.39$ (opens)
over all pivotal points	$24/114 = 0.21$ (opens)	$15/114 = \mathbf{0.13}$ (closes)

Table 3: The V2 gate (threshold 0.20) under the four accountings the registration left open. It opens under three and closes under the strictest.

formatting 0.667 ($n = 12$), termination 0.100 ($n = 10$), test scheduling 0.333 ($n = 3$); per-type rates are descriptive (point classification draw-stable at only 17/23; verdict and gate survive re-draws). Interaction is nonzero at 29/48 points with $|I|$ up to 0.73 — on this stack interaction is the dominant component of harness credit, not a correction to it.

The gate, under every accounting. The registration fixed the threshold (non-screened mass ≥ 0.20) but left two accounting choices under-specified — the denominator (measured points only, or all pivotal points) and whether the 10 analytically degenerate termination points (see last-mover screening below) count as evidence — and the verdict turns on both (Table 3).

We report all four rather than adjudicate post hoc; the denominator and the treatment of degenerate points must themselves be registered. Robust across accountings: substantial non-screened pivotal mass exists (15 points with $|I|$ up to 0.73, vs. zero V1 slack breaks), richer stacks de-screen as the anatomy predicts, and the training experiment is licensed under the measured-points accounting.

The licensed branch, exercised. The licensed training ran (pre-regs 32–33; four arms, 3 seeds, dose- and episode-matched). The first run aborted at calibration: the V1 cost-weight grid is mis-scaled to V2 token magnitudes (λ does not transport). Re-registered ($\lambda^* = 0.2$, 10 margin-verified tasks), it returned the declared outcome s3, attractor collapse: every learning arm collapses onto a fixed attractor — 8/9 runs token-exactly onto summarize-always (held-out R_{eff} 0.831 vs. 0.889 for the dominant expressible target; one outcome seed and the untrained control at keep-always, -0.092) — before any credit ordering becomes testable. A registered probe (pre-reg 34) locates the failure in the landscape, not the optimizer: at 5$\times$ smaller learning rate and 2$\times$ budget both arms still collapse token-exactly ($\theta \approx 0$, opposite attractors), and no λ yields even 3 tasks with attractor margin ≥ 0.10 — the pool a power-gated re-run requires does not exist on this stack. Both branches end in identified negatives: credit cannot pay for its replays (closed); the optimization collapses before credit is tested (open) — each predicted by pre-training measurement.

Last-mover screening. Terminal flips (continue$\rightarrow$terminate) make the joint arm analytically degenerate: the counterfactual episode ends at the harness decision, the model's alternative never executes, and $R_{HM} \equiv R_H$ by determinism (addendum 29a). These 10 termination points mirror the central finding: here the harness screens off the model. Screening-off is a fact about which layer moves last across the channel that reaches the reward.

Scope. Three caveats bound the verdict. Power and selection: $n = 48$ with 65 points excluded for want of an a' makes per-type rates wide, and the exclusion selects on model entropy — though the cut runs both ways: at excluded points the model is near-deterministic, so the correction is vacuous there by construction; test scheduling ($n = 3$) supports no claim of its own, and restricting to types with $n \geq 10$ leaves s3 unchanged (context 14/23, observation 8/12 screened, both < 0.75). Estimand: the pre-registered controls (pre-reg 30, Appendix E) discipline the verdict after the fact: it survives both a' re-draw schedules, and under the information-poor a'_s the de-screened points turn exactly additive ($I = 0$) — so the V1-closed / V2-open contrast still compares verdicts under estimands whose completeness differs. Addenda: two protocol amendments (29a, 29b) were registered mid-run — before analysis, after data contact — and the verdict inherits that weaker label where they bind.

8 Threats to Validity and Discussion

8.1 External validity

Our tasks are designed, not sampled from a public benchmark — a motivated limitation: the measurement needs control of destroyable-information position, calibrated per-stratum recoverability, and deterministic fractional reward, which no public benchmark provides jointly. We ran the missing experiment in reduced form on MBPP+ and HumanEval+ (pre-regs 16/20/25): the floor replicates exactly in every cell (3,668 nulls) and screening replicates unconditionally (Table 2), while every pressure cell saturates away, so the regime contrast stays untested externally (Appendix D); the phenomenon is no task-design artifact.

8.2 Replication across models and pools (summary)

Appendix D gives the full record: Qwen3-1.7B, Qwen3-8B, a pool hardened to the 8B's competence, and the pre-registered symmetric control. The floor replicates everywhere; screening replicates unconditionally on the 1.7B and both external environments; the 8B breaks the shield at a flat rate in both regimes; the 4B also breaks on the curated pool. The claim re-scopes: whether screening holds is a property of the (model, harness, task) triple, with break sets point-level disjoint across models, and the re-injection mechanism is neither necessary nor sufficient there — all Qwen3-family; cross-family transport unmeasured.

8.3 Statistical caveats

Clustered inference. Points within a task are not independent. The pressure-vs-slack contrast is 3 tasks vs. 0 at task level (sign test $p = 0.125$) and $p = 0.005$ at point level. We report all levels and rest the screening claim on the task-clustered estimate (53/56, CI95 $[0.87, 1.00]$; 19/22 on unique points) together with cross-configuration break recurrence — not on the sign-test p-values (they certify only non-coin-flip; Appendix C).

Multiplicity and post-treatment conditioning. Only pre-registered outcomes (Table 7) are confirmatory; borderline point-level values are descriptive, and saturation-conditioned rates condition on a post-treatment variable.

8.4 Estimand caveats and the practical prescription

Appendix E details the estimand-level caveats: the shared sampled alternative coupling C_M and I; the single-draw a', bracketed on both stacks by re-sampling and a'_s studies (V1 99/101, 24/31; V2 pre-reg 30, §7); selection into measurement (53–73% discarded, composition pointing against the headline contrast); the 5.6× episode tax; and the stochastic protocol (the census predicts where the temperature-0.8 estimand leaves its null band).

The practical prescription. No fixed correction transports across stacks; the procedure does: null floor, screening census, a power gate (train only where verified attractor margins resolve at the planned budget — it vetoed our V2 round, pre-reg 34), then decide which signal is billable, plus the identity $C_H = -I$ at screened points and the correction $C_{HM} - C_M$.

9 Conclusion

We built the first apparatus measuring harness credit, model credit, and their interaction on the same trajectory against an exactly-zero floor: with slack the model screens off the harness (57/57) and single-layer harness credit double-counts; under pressure the shield breaks — a triple-dependent contrast. Better credit did not train a better harness on either stack. Contributions: the infrastructure, the triple-dependence finding, the APC warning, the census and power gates, and a 34-item ledger whose training negatives are the design brief for the missing benchmark (Appendix L).

A Master Claim Table

Table 4 maps every claim of the paper to its evidence and epistemic status.

B Screening-Off, Formally

The measurements above are point identities, not expectations: under the premises of §4.1 (greedy decoding, fixed config, sequential serving, prefix caching off — with the resulting exactness measured, floor 0.0), the model, the environment, and the harness are all deterministic, so each replay branch is a single deterministic rollout. Fix a recorded trajectory with reward R_0 and a decision pair at index t: harness action h with alternative h', model action a with sampled alternative a'. Writing $R_H = R(\mathrm{do}(h'))$, $R_M = R(\mathrm{do}(a'))$, $R_{HM} = R(\mathrm{do}(h', a'))$ for the replayed rewards, the estimands are $C_H = R_0 - R_H$, $C_M = R_0 - R_M$, $C_{HM} = R_0 - R_{HM}$, and $I = C_{HM} - C_H - C_M$; throughout this section R is the test-fraction reward (not R_{eff}, whose token term accumulates over the differing prefixes), matching all measurements of §4.3. Screening-off at the point means $C_{HM} = C_M$.

Proposition 1 (Double-counting identity). At any point, $C_{HM} - C_M = C_H + I$. Consequently, where screening-off holds: (i) the marginal credit of the harness flip in the presence of the model flip is zero, $C_{HM} - C_M = 0$; (ii) the single-layer harness credit equals the negated interaction, $C_H = -I$; (iii) a trainer that bills the harness C_H at such a point therefore assigns a signal consisting entirely of interaction — credit for an effect the model absorbs — while the interaction-corrected signal $C_{HM} - C_M$ vanishes.

Proof. The first identity is the definition of I rearranged. Under $C_{HM} = C_M$, it gives $0 = C_H + I$; (i) and (iii) follow by definition. □

Proposition 1 upgrades the empirical observation to a correction rule, with an honest scope note. C_H and $C_{HM} - C_M$ are the two orderings of a two-player marginal-contribution calculation, and which one a trainer should bill is a modeling choice tied to the training configuration: C_H is the advantage against a fixed model policy (the model replays its factual actions and adapts downstream), appropriate when only the harness trains; $C_{HM} - C_M$ is the harness's marginal contribution in the world where the model deviates, the relevant wedge for joint training, where billing the harness for effects the model's own update will absorb double-charges the same reward; a Shapley symmetrization would bill the average, $C_H + I/2$. Our contribution is not to legislate the ordering but to measure the wedge between them — which is exactly I, and which our census shows is (a) usually the whole of C_H at screened points and (b) obtainable for one extra replay. Our training arm 3 uses $C_{HM} - C_M$; arm 2 uses C_H; the comparison never separated (§6), so the practical import of the choice remains empirically open — what is settled is its size.

Remark 1 (Additivity signature). The dual of screening is additivity: $I = 0$ with $C_H \neq 0$, i.e. $R_M - R_{HM} = C_H$ — the layers' effects compose independently and single-layer harness credit is exactly correct (no double-counting). Which identity holds at a point is a property of the estimand of a', not of the point alone: on the V2 stack, replacing the sampled alternative a' (drawn in the factual, information-rich state) with a'_s (the same sampler run in the counterfactual, post-summarization state) moves 13 of 13 non-screened points to exact additivity ($I_s = 0.0$, $C_H \neq 0$; pre-reg 30, Appendix E) — whereas the V1 de-screening under a'_s showed a re-injection signature instead. A census therefore licenses credit only under the estimand it was run with: the identity structure ($C_H = -I$ vs. $I = 0$) can flip with the estimand while the trajectory stays fixed.

Lemma 1 (Coalescence). Let $s_k = (c_k, e_k)$ denote the agent state at step $k \geq t$ — model-visible context and environment (sandbox) state — and consider the two branches $B_M = \mathrm{do}(a')$ and $B_{HM} = \mathrm{do}(h', a')$. If the agent is deterministic and there exists $k \geq t$ with $s_k^{B_M} = s_k^{B_{HM}}$, then the branches coincide from k onward; in particular $R_M = R_{HM}$ and screening-off holds exactly.

Proof. The one-step transition (model call, tool execution, harness policy) is a deterministic function of s_k. Equal states yield equal actions and equal successor states; induction to termination yields identical suffixes, hence identical final rewards. □

Corollary 1 (Behavioral coalescence at the intervened turn). If both branches execute identical action sequences from turn t to termination, then $R_M = R_{HM}$. (Harness context operations do not modify the environment, so the environment states at turn t agree; identical action suffixes applied to equal environment states under a deterministic environment yield equal final environment states, and R is a function of the final environment state.)

The two empirical mechanisms of the anatomy are the two coalescence-inducing events we observe in practice; the lemma's role is to guarantee that whenever coalescence occurs, the resulting equality is exact rather than approximate. Because reward equality does not certify coalescence (with reward resolution $1/n_{\text{tests}}$, collisions are possible), we verified the hypotheses directly in the recorded branch traces of all 75 screened points, comparing per-turn states (c_k, e_k) and per-turn behavior (action, observation), normalizing non-semantic identifiers that leak into test output (memory addresses, sandbox paths). The result, stated so that each count matches an enunciated argument: in 29/75 points the full state (c_k, e_k) becomes byte-identical within 1–4 turns of the intervention — Lemma 1's hypothesis, verified; in 72/75 (a superset of the 29) the two branches execute identical action suffixes from the intervened turn itself — Corollary 1 applies and entails $R_M = R_{HM}$. The formal arguments therefore explain 72 of the 75 measured screening events. The 3 residuals: two (l_log_parser idx 10, in two configurations) are exactly the reward-collision case — $R_M = R_{HM}$ holds without trace coalescence, so the screening statistic counts them but the theory does not explain them — and one (l_payroll_pipeline idx 7 under g900, saturated at $R = 1.0$) coalesces behaviorally only 3 turns after the intervention, which is consistent with screening but not covered by Corollary 1; we declare all three rather than absorb them into the explained count. No suffix violated determinism: whenever states matched at some turn, the branches were identical from that turn to termination, corroborating the premises at trace level. Re-triggering (22/52): both branches executed the same a', so environment states already agree; the live harness in B_M summarizes at step $t{+}1$, and context coalescence follows. Re-injection (23/24 annotated): a' (typically a write_file carrying the destroyed constants) equalizes the environment component, and the residual context difference is confined to content the greedy continuation never conditions on. The lemma thus explains a striking feature of the data: measured screening-off consists of exact zeros, not small values — determinism converts coalescence into equality, the measured 0.0 floor makes equality observable, and the coalescence itself is now measured rather than presumed.

Remark 2 (Pivotality and the competence window). Call a decision point reward-pivotal if at least one of the three interventions changes the reward. Non-pivotal points satisfy $C_H = C_M = C_{HM} = I = 0$ identically: nothing is detectable there, by construction rather than by noise. When the agent fails a task outright and no intervention — single or joint — rescues it, every point on that trajectory is non-pivotal. This is the formal content of the competence window: reward saturation does not merely add variance — it removes pivotal points, which is why the Qwen3-1.7B pressure regime (11/12 saturated) had essentially no power and why the mt4 break rate collapses.

Corollary 2 (What single-layer training optimizes). In a regime where screening-off holds at all measured points (our slack regime, 57/57), the per-decision signal of a single-layer harness trainer is $-I$ at every shielded point: it consists entirely of credit for effects the model absorbs. More generally, by Proposition 1 the single-layer signal errs by exactly I at every point where $I \neq 0$, regardless of regime — at the broken point l_vending_machine, $C_H = -0.09$ while the corrected signal is $+0.09$, an inverted sign. The corrected signal $C_{HM} - C_M$ is exact in both regimes.

Screening-off has a direct budget consequence for any attempt to train on replay credit — one that is decidable before training begins. We state it here as the operational centerpiece of the paper; §6 then tests it the hard way.

Proposition 2 (No free lunch for replay credit under screening). Fix a task pool, the determinism premises of §4.1, and a training budget matched in total LLM calls. Let σ be the fraction of reward-pivotal harness points at which screening-off holds exactly, and consider a trainer that bills the harness the corrected signal $C_{HM} - C_M$, paying $r \geq 2$ replays per billed point out of the shared budget. Then: (i) every screened point contributes a billed signal of exactly zero (Proposition 1(i)) at strictly positive replay cost; (ii) the total billed per-decision signal is confined to the $(1-\sigma)$ non-screened pivotal mass and bounded by the effect sizes $|C_{HM} - C_M|$ there — so any training value that flows through the billed signal is bounded by (non-screened mass × effect size) minus the replay tax; (iii) if $\sigma = 1$ on the measured pool, the consequence depends on the update rule, and both cases are dominated: a trainer that bills only the corrected credit receives an identically zero gradient and reduces to the untrained policy — which is exactly what Act 4 measures (the corrected arm lands on the untrained control, 0.392) — while a trainer that mixes an outcome term with the (zero) credit term reduces to outcome-only policy gradient computed from a strictly smaller episode budget.

Proof. (i) is Proposition 1(i) plus the accounting of dose-matching. (ii) sums (i) over points: screened points contribute zero billed signal, non-screened points at most their effect size, and every replay is paid from the shared call budget. (iii) case one: with the billed credit identically zero, every per-decision update vanishes and the policy stays at initialization; case two: with the credit term zero, the update rule coincides with outcome-only on the episodes it can still afford. □

The bound in (ii) prices the billed signal, not every conceivable channel of value: a correctly-zero credit could in principle still help as a variance-reduction baseline against the noisy outcome return assigned to the same points. We state this channel explicitly rather than bound it; on our stack Act 4 shows no such benefit (the corrected arm does not outperform outcome-only anywhere), but the proposition does not exclude it elsewhere.

Corollary 3 (The census is a decision procedure). Both quantities in the bound — σ and the non-screened effect mass — are measurable before any training by a screening census whose cost is a small multiple of baseline collection. Whether cross-layer replay credit can pay for itself on a given stack is therefore decidable in advance: pre-register a gate on the census and run credit training only if it opens.

Three scope notes keep the proposition honest. First, the dominance in (iii) is a property of the measured pool, not of the method class: pools with substantial non-screened pivotal mass may repay credit — the proposition prices replay credit, it does not prohibit it. Second, the tax term attaches to the replay estimator under dose-matching, not to the signal itself: an amortized critic fit offline to census data moves the replay cost out of the training loop and changes the accounting — measurement remains the right place for replays, and the census is exactly the dataset such a critic needs. Third, screening annihilates the harness-side signal only: C_M is untouched, so model-side credit is priced by a different budget line. Under this reading, Act 4's null is not an optimization pathology but the correct behavior of an exact estimator of a quantity that is truly zero — the arms could not have separated in any other direction.

We emphasize the epistemic status of each piece. Proposition 1 is elementary algebra — its value is not mathematical depth but the correction rule it licenses. Lemma 1 holds given the determinism premises, and both the premises (§4.1) and the lemma's hypothesis (this section) are measured rather than assumed. Whether coalescence occurs at a given point, which mechanism induces it, and how often screening-off holds at all are empirical questions that the rest of this section answered.

B.1 Control tasks and the c_temp_label exception

On control tasks, where summaries destroy nothing by construction, the method returns $C_H = 0$ in 11/12 measured points — the construct behaves. The exception is informative: c_temp_label showed $C = +0.22$ at turn 0 despite all task-critical information surviving the summary. Audit of the replay pair shows the effect is behavioral, not informational: the summarized context changes the model's subsequent formatting behavior, which a strict test then penalizes. C_H measures total causal effect through any pathway; the controls certify the informational pathway is clean, and the exception demonstrates the behavioral pathway exists and is detectable.

C Shield Anatomy and Robustness

Anatomy of the shield. The sets in play: across the three slack configurations and the first pressure configuration we measured 78 interaction points (57 slack + 21 at mt6); 75/78 show exact screening-off ($C_{HM} = C_M$) — the 3 exceptions are the mt6 pressure breaks. We report this estimation-first, on the falsifiable subpopulation: by Remark 2, reward-non-pivotal points satisfy $C_{HM} = C_M$ by construction, so the informative unit is the 56 reward-pivotal points. There the screened fraction is **53/56** = 0.946, CI95 $[0.87, 1.00]$ by task-clustered bootstrap over the 16 pivotal tasks; collapsing further to unique task×point pairs (removing cross-configuration pseudo-replication) gives 19/22, and 13 of 16 pivotal tasks screen at every measured point. Binomial sign tests against a 50/50 null give $P = 4.1 \times 10^{-13}$ (instances) and $P = 4.3 \times 10^{-4}$ (unique pairs), but we do not rest the claim on them: no substantive model of the data-generating process fixes the non-screening probability at 0.5, so these p-values certify only that the pattern is not coin-flip noise. The inference we retain is the clustered estimate; a bundled script (estatisticas_pivotais.py) re-derives every number in this paragraph from the raw artifacts and fails on drift. One further split the timeline forces on us: the screening-off hypothesis was formulated after seeing the first slack census (g600, whose artifacts predate the registration commit by 34 minutes), so g600 is discovery data. Splitting accordingly: discovery (g600) screens 14/14 pivotal points; the confirmation set (g450/g900/mt6; g450 was in flight at registration, g900 and mt6 strictly subsequent) screens 39/42; restricting to the two strictly post-registration configurations (g900+mt6) gives 27/30. The estimate does not depend on the discovery cell.

Robustness to the alternative draw (pre-registration 21). The sharpest objection to every number above is that C_M, $C(HM)$, and hence screening are relative to a single sampled a' per point. We pre-registered outcomes (stable ≥ 0.90 / partial / fragile < 0.75, over informative slack draws) and re-sampled a' at all 56 pivotal instances under three disjoint seed schedules (3001–3008, 4001–4008, 5001–5008; same procedure, temperature 0.8), yielding 152 new draws (16 no-alternative outcomes; 304 extra replays), of which 81% differ from the published a'. The pre-declared stable outcome was met: 99/101 informative slack draws screen exactly (rate 0.980, CI95 task-clustered $[0.944, 1.00]$); restricted to genuinely new alternatives, 79/81. The two non-screened draws are themselves informative: one is a single-draw excursion at the already-declared reward-collision residual (l_log_parser idx 10[1]), and one is a genuine synergy break at invoice_pricing under one of its three draws ($I = +1.57$; the other two draws screen) — so screening at a point is a dominant-but-not-sure property of the alternative distribution, not a logical constant. Symmetrically, the three published mt6 break points break again under every redraw (9/9): breaks are properties of the decision point, not of the draw. Sixteen draws found no alternative (the selection channel of §8, rate consistent across schedules: 6/4/6). Among the screened points, 52 are non-trivially shielded: the harness flip alone has a real effect ($C_H \neq 0$) that the joint flip erases — these are exactly the double-counting cases. Of the 52 shielded points, 24 carry constant-dictionary annotations that make the re-injection check possible (the remaining 28 are unannotated V2 points, which we declare rather than extrapolate). Among the annotated: 23/24 show structural re-injection — the sampled alternative action a' carries the destroyed information across the cut, a property of the estimand (since a' is sampled from the model conditioned on the un-summarized state); 9 also show the second mechanism and 1 shows neither. The second mechanism is checkable on all 52 shielded points: in 22/52 the live downstream harness re-triggers summarization anyway, equalizing the branches. Section B makes this division precise: a coalescence lemma and a one-line corollary guarantee that whenever the two branches coalesce, the resulting equality is exact — and we verify the occurrence directly in the recorded traces (the arguments explain 72/75 screened points; the 3 residuals, including 2 reward collisions, are declared). We also found genuine

[1]Point labels throughout are checkpoint indices (cp_index); raw artifacts also carry the 1-based tool-call index (= cp_index + 1), so this point appears as index 11 in results.jsonl.

non-saturated synergy where the shield breaks (l_vending_machine: $C_H = C_M = -0.09$, $C_{HM} = 0$, so $I = +0.18$: either flip alone hurts, together they cancel).

D Replication Across Models, Pools, and Environments

Floor and unconditional replications. The floor replicates everywhere (pre-regs 14, 15, 16, all registered before running): the 8B contributes 810 dedicated and 234 attached null replays and MBPP+ contributes 1,294, all exact. Screening-off replicates unconditionally on Qwen3-1.7B (0 raw breaks in 23 points) and on MBPP+ (66/66 raw; 11/11 pivotal).

The 8B breaks the shield. On Qwen3-8B it does not: raw break rates are flat across regimes (5/28 slack, 6/34 mt6, 7/35 mt4), so the slack/pressure contrast established on Qwen3-4B does not transfer to the 8B, and we report this as the headline of the replication rather than conditioning it away. Saturation-based exclusion would be invalid here: clipping R at $\{0, 1\}$ can only shrink $|C_{HM} - C_M|$, so saturation can mask nulls but cannot fabricate breaks — a saturated break of 0.9 is a real break. The 8B slack breaks have a sharp common structure ($C_H = 1.0$, $C_M = 0.0$, $C_{HM} \in [0.8, 0.9]$, hence $I \in [-0.2, -0.1]$): the harness flip alone is catastrophic, the model flip alone is null, and the joint flip tracks the harness — the shield inverts. The mechanism is the one the anatomy of §4.3 identified: on the same tasks the 4B's sampled a' re-injects all destroyed critical constants (5/5, 5/5, 6/6 on the shielded points), while the 8B's a' re-injects none (0/5, 0/6, 0/8, 0/5; matching by constant-literal substring in the serialized a') — its solutions are structured so that the alternative action no longer carries the constants. The mechanism is verifiable on 4 of the 5 slack breaks (the fifth, invoice_pricing, has no annotated constants), and the breaks recur: all 5 slack break tasks break again, with identical per-task structure, in both pressure configurations (6/34 at mt6, 7/35 at mt4). Whether screening-off holds is therefore itself competence-dependent: at that stage of the design the shield appeared to be a property of the model–harness pair — a claim the symmetric control below narrows further to the (model, harness, task) triple (the closest precedent, model-dependent counterfactual measurability (Zhang, 2026), concerns whether ground truth can be estimated at all; here the recovered causal structure itself inverts). We say competence, not scale: 4B and 8B differ in post-training as well as size, and the claim is anchored in the measured re-injection contrast on the original pool (which, as reported below, does not generalize as a mechanism), not in parameter count. Under pressure the 8B also shows a genuine non-saturated synergy break at l_door_controller ($C_{HM} = 0.23$ vs. $C_M = -0.46$, $I = +0.38$). Saturation still dominates both auxiliary models in opposite directions (11/12 of the 1.7B's pressure points, from weakness; 90/97 of the 8B's points, from strength — and the slack regime leaves the 8B a single non-saturated point, so the conditioned slack contrast has essentially no power and we do not claim it).

Curated pool (pre-registrations 17–18). The remaining step — the break rate on a pool curated to the 8B's competence — was then run (pre-registrations 17 and 18): a first hardened pool failed its pre-registered curation rule (17/24 tasks solved at 1.0; reported as failure), and a second pool targeted at the fractional-reward profile passed (22 of 44 tasks in the $(0.05, 0.95)$ window; 792/792 nulls exact after the ASLR fix of §4.1). In-window, the 8B's raw screening breaks replicate and stay flat across regimes — 3/27 slack, 3/31 pressure — including a non-saturated slack break (x_hours_bank: $C_H = +0.25$, $C_M = -0.17$, $C_{HM} = 0.00$, $I = -0.08$), which closes the clipping objection: breaks persist where saturation cannot manufacture them. The pool-matched competence contrast is the 8B replication's (pre-reg 15; same 30 tasks: 4B 0/57 raw vs. 8B 5/28); within the curated pool the regime contrast seen on the 4B does not exist on the 8B even inside its competence window. Two of the three break points recur across both regimes with identical structure. Two pre-registered sub-hypotheses failed: the binary re-injection mechanism does not generalize (at active slack points the curated-pool a' never re-injects constants, 0/15, yet only 3/15 break; under pressure 2 of 3 breaks occur with re-injection; one-sided Fisher $p = 1.0$ and $p = 0.94$ — the slack table is degenerate, no exposed points, so only the pressure test is informative, and it points the wrong way), and the directional prediction (pressure $\geq$ slack) failed (9.7% vs. 11.1%). All curated-pool break counts are descriptive: with 3 break tasks per regime we make no clustered inferential claim, and we make no cross-pool p-value comparison against the 4B's 0/57.

Symmetric control (pre-registration 19) and the triple claim. Two design threats were named at this point — the curated comparison was cross-pool (the 4B's 0/57 comes from the original pool — the paired contrast on identical tasks is pre-reg 15's 0/57 vs. 5/28); and the curation window is operationalized by task family (the second pool was designed around the profile that survived curation), so in-window break rate could be a property of the family rather than of competence — and the pre-registered symmetric control (pre-reg 19: the 4B on the same 22 curated tasks, outcomes declared in advance) resolved both, against the strong form of our own claim. The 4B also breaks raw screening on the curated pool: 4/16 slack points (3 non-saturated, including a synergy break at h_turnstile_fsm with $I = +0.77$ and h_hotel_folio at $I = -1.42$) and 2/23 pressure points; its floor held (396/396 nulls exact), and the pivotality-based prediction that the 4B would sit saturated-in-failure on this pool failed (1/16 and 3/23 saturated — reported as a failed pre-registered sub-hypothesis). The competence claim therefore re-scopes: the shield is a property of the (model, harness, task) triple, not of the model–harness pair alone.

What survives every cell. The pool-matched model contrast on identical tasks (0/57 vs. 5/28), the majority of points screened in every configuration (worst cell 12/16), the regime contrast existing only for the 4B on its original pool, and a majority pattern of break-point stability: break sets are disjoint across models at the point level (the two models share one break task, h_hotel_folio, at different turns), and within a model most break points recur across regimes — in the 8B, 2 of 3 slack break tasks recur numerically identically; in the 4B, 2 of 3 break tasks recur, one numerically identically (h_hotel_folio), one with the same sign structure but different magnitudes (h_turnstile_fsm: $I = +0.77$ slack vs. $+0.31$ pressure), and h_telco_roaming does not recur. A further reading — breaks concentrate where the pool sits near the model's competence boundary (for the 4B, the original pool under pressure and the hardened pool in both regimes; for the 8B, the hardened pool in both regimes) — is descriptive and not pre-registered: the pre-declared interpretation of outcome c2 was "a property of the task family, not of competence," and what favors the triple reading over the pure-family reading is the point-level disjunction of break sets between models on the identical pool. The break structure is also heterogeneous — the original-pool 8B slack signature ($C_H = 1.0$, $C_M = 0.0$, $C_{HM} \approx 0.9$) does not recur — and a descriptive trace anatomy of all 8B and 4B curated-pool breaks (0 new rollouts) finds one uniform base, three recurring patterns, and a model asymmetry (restore-direction flips appear only in the 4B); Appendix K reports it in full, including two audit details showing that recurring break points are numerically identical across regimes while I magnitudes are baseline-relative. The robust claim is therefore narrower than the mechanism story: whether screening-off holds is a property of the (model, harness, task) triple, and re-injection of destroyed information, which fully explained the annotated 4B/8B contrast on the original pool (pre-reg 15), is neither necessary nor sufficient for the shield in a curated pool.

External environments in full (pre-regs 16, 20, 25). We therefore ran the missing experiment in reduced form (pre-reg 16, registered before running): a multi-turn MBPP+ adaptation — 60 tasks derived mechanically from MBPP+ (Liu et al., 2023) with fractional reward (6–10 test functions per task, canonical solutions validated in our sandbox), one slack and one pressure configuration on Qwen3-4B. The floor replicates exactly (1,294 null replays, all exact) and screening-off replicates at 66/66 interaction points, of which 11 are reward-pivotal (11/11 screened) — the other 55 screen by construction (Remark 2), so the informative content of the replication is the pivotal 11/11, not the 66/66 — including a non-saturated point with $I = -0.4$, a live double-counting case in an external benchmark. What the adaptation cannot yet test is the pressure-regime break rate: 60/66 points are reward-saturated (Qwen3-4B is too strong for MBPP+), leaving 6 non-saturated points and no breaks — inconclusive at that power, the same saturation boundary seen at 1.7B and 8B. A pre-registered second attempt (pre-reg 20: the 1.7B on the same MBPP+ pool, declared outcomes) landed on the same boundary from the same side: floor exact (1,080/1,080), screening replicates raw at 44/44 points across both regimes (24/24 reward-pivotal) — a second (model, environment) cell with unconditional screening in the external benchmark

— but 20/22 points per configuration are saturated (the 1.7B still solves 34/60 MBPP+ tasks at 1.0) and the pressure budget never binds on MBPP+'s short trajectories, so the pressure cell of environment B remains empty after two attempts, and we declare the cell empty. A third environment (pre-registered as pre-reg 25 with the same frozen mechanical derivation, from HumanEval+ (Liu et al., 2023): 60 tasks, 164 processed, zero discarded) reproduces the pattern a third time: floor exact (1,294/1,294 nulls) and screening at 76/76 interaction points, of which 42 are reward-pivotal (42/42 screened), with the pressure cell again saturated away (2 non-saturated points) — declared inconclusive by the pre-specified power gate.

E Estimand and Infrastructure Caveats

Multiplicity. This paper reports on the order of a dozen p-values across regimes, strata, and clustering levels, and we apply no family-wise correction. Only the pre-registered outcomes (Table 7) are confirmatory; borderline point-level values ($p = 0.037$, $p = 0.005$) would not all survive a family correction, and we treat them as descriptive support. The claims we retain rest on replicated structure — exact screening at pivotal points, cross-configuration recurrence of break points, and the pool-matched model contrast — rather than on any single threshold crossing.

Saturation is endogenous, and conditioning on it is post-treatment. Reward resolution is set by test count, so saturation is partly a design artifact. Worse for inference: saturation is defined on the rewards of the replayed arms, i.e. on outcomes of the intervention itself, so every saturation-conditioned rate in this paper conditions on a post-treatment variable and carries selection bias of unknown sign. We therefore use conditioned contrasts descriptively only, never confirmatorily, and we describe saturation as a correlate of the competence window, not a mediator. The inverted-U at mt4 is only partially accounted for (conditioned rate 0.045 $\rightarrow$ 0.143; the ordering mt8 > mt6 > mt4 persists), so the window remains a partially supported hypothesis, discussed but not claimed as a contribution.

Estimator coupling. C_M and I share the sampled alternative action a': they are not independent evidence, and the structural component of screening-off (§4.3) is partly a property of the estimand — a' sampled from the un-summarized state can re-inject information. We quantified this (23/24 annotated shielded points; 28 unannotated points declared) rather than hiding it; an estimand with a' sampled from the summarized state would measure a different, also legitimate, quantity. Two further limitations of the estimand: a' is a single sample per point (fixed seed schedule, §3) in the census; the pre-registered re-sampling study (§4.3) measures sensitivity to that draw directly and finds the slack result stable (99/101) and the pressure breaks draw-invariant (9/9). What remains unmeasured is coverage of the alternative distribution beyond $k = 3$ redraws at temperature 0.8. The alternative estimand — a'_s conditioned on the summarized state, so that re-injection is impossible by construction — we ran as a registered follow-up (declared outcomes; all 44 keep$\rightarrow$summarize pivotal instances, one seed schedule, 2 replays each, zero sampling failures, all 44 a'_s distinct from the published a'). The result splits cleanly: screening under a'_s holds at 24/31 informative slack points (0.774, task-clustered CI95 $[0.69, 0.88]$ — the pre-declared "partial" bin, down from 53/56 under a'), and all seven de-screened points have the same signature: $C_M(a'_s) = 0$ and $C_{HM}(a'_s) \approx C_H$ — exactly what turning off the re-injection channel predicts. The two estimands therefore bracket the phenomenon: with an information-complete alternative the model absorbs essentially everything (53/56); with an information-poor alternative the harness flip's damage reappears in the joint branch at about a quarter of the points. Screening is real in both estimands but its completeness is partly a property of the alternative's information access — the headline claim is conditional on the estimand, and we scope it so. Under pressure the picture inverts symmetrically: both published mt6 break points in this direction (l_log_parser idx 7, api_router idx 0) screen under a'_s — so break-ness, too, is a property of the (point, alternative) pair rather than of the point alone, consistent with the stochastic cell where the weakest break dissolved under re-sampling. (The third published break is a restore-direction flip, outside this estimand's population by construction.)

V2 estimand controls (pre-reg 30). The same two controls, run on the V2 census's 48 measured points. Re-sampling (two disjoint seed schedules at each point's census temperature): the plain-text sampler is near-deterministic even re-seeded (56/96 draws return the identical a'; 17 find none), and on the 23 informative draws the point-level classification is draw-sensitive — stability $17/23 = 0.739$, with flips in both directions across three decision types. The aggregate verdict is not: substituting each schedule's draws and recomputing, the census outcome remains s3 and the primary gate stays open under both schedules (0.474 and 0.368 vs. 0.395 original; threshold 0.20). Summarized-state a'_s (23 keep→summarize context points, one schedule): screening drops to $8/21 = 0.381$ — and all 13 non-screened points (9 genuinely de-screened) share a signature different from V1's: $I(a'_s) = 0$ exactly with $C_H \neq 0$, i.e. the layers become additive — the one regime where single-layer credit is correct. Which identity holds at a point — screening ($C_H = -I$), additivity ($I = 0$), or neither — is itself a function of the estimand, which sharpens the prescription: run the census under the estimand you intend to bill. Selection composition, V2: the 66 pivotal points excluded for lack of an a' have flip effects comparable to the measured 48 (median $|\Delta R|$ 0.167 vs. 0.182) but sit earlier (median turn 2 vs. 7) and in smaller contexts (919 vs. 2,099 tokens), with termination and observation over-represented and both retry points excluded — the census over-represents late, large-context states where the sampler has entropy, a direction-neutral but real selection.

Selection into measurement. A point enters the interaction census only if the sampler finds a parseable $a' \neq a$ within the 8-seed schedule; across configurations 53–73% of candidate points are discarded this way. The measured population is therefore selected on the model's own output entropy at that state — a post-decision variable plausibly correlated with both competence and the information content of the context — and every break or screening rate is conditional on "an alternative exists." For the headline cross-model contrasts the discard rates are close enough that differential selection is an implausible driver: on the pool-matched contrast (identical 30 tasks) the 4B discards 66–73% across its slack configurations versus 64% for the 8B at the same budget, a ≤9-point selection gap against a 0% vs. 18% raw break-rate gap; on the curated pool the rates are 57–70% (4B) versus 53–59% (8B). We measured the composition directly (analise_selecao.py, from the per-point sampling logs joined to the baseline trajectories): selection is real and directional — in both models the retained points sit earlier in the trajectory (pool-matched relative position, found vs. discarded: 0.33 vs. 0.73 for the 4B, 0.21 vs. 0.79 for the 8B; task-clustered bootstrap CIs exclude zero) and carry smaller contexts. But the direction disfavors the headline contrast rather than explaining it: the 8B's retained points are shallower than the 4B's (mean turn 0.65 vs. 1.55, context 395 vs. 433 tokens, $d = 0.31$–0.54), so if depth promotes breaks, composition biases the 8B toward fewer measured breaks — and it still breaks at 18% against the 4B's 0%. What selection does forbid is any claim about deep-context interaction: no census measures points the sampler cannot reach, and the discarded majority is precisely the late-trajectory, large-context population.

Serving premises. The exact floor holds under a fully specified stack: vLLM 0.8.5.post1 (V1 engine, pinned; PyTorch 2.6.0+cu124, CUDA 12.4, driver 580.173.02, Python 3.11.14, full transitive closure pinned in the repository's uv.lock) serving one model per experiment on a single NVIDIA RTX 4090; dtype auto (bfloat16); CUDA graphs enabled (no enforce_eager); default chunked prefill (max_num_batched_tokens = 2048); automatic prefix caching explicitly disabled; GPU memory utilization 0.85 (0.92 for the 8B); strictly sequential requests against a single instance; greedy decoding (temperature 0, single seed) for every episode and replay rollout; and the fixed a' seed schedule of §3. The floor is a measured property of this stack, not a theorem; any deviation (batching, caching, config drift, a version bump) voids it and must be re-measured.

Protocol cost. Exactness is bought with throughput: the premises force strictly sequential serving, one request at a time, on one GPU. The null-replay inventory is the canonical 9,296 of Table 8 (dedicated plus attached, reconciled per artifact); the counterfactual inventory adds 1,380 replays at 460 interaction points (three per point), 13 census points and 180 stochastic replays in the round-4 cells, 294 single-flip C_H replays, 304 re-sampling replays with 401 a' sampling calls (pre-registration 21), and 1,617 census a' sampling calls (up to 8 seeds each); the V2 census chain (§7) adds 120 episodes, 117 nulls, 693 screening replays and the census arms in roughly 75 GPU-minutes; the training comparison adds 4,804 episodes (Act 2) plus the dose-matched replays. The chained experiment logs account for roughly 25 wall-clock hours on the single RTX 4090 (an under-count: the earliest runs predate chained logging). Dose-matching makes the replay tax concrete: at equal rollout budget, the interaction-corrected arm afforded 319 training episodes against the outcome-only arm's 1,789 — a 5.6× episode cost that any replay-based trainer must amortize.

Beyond determinism: the stochastic protocol. Real agent stacks violate every exactness premise (sampling temperature, flaky tests, batched serving). The protocol degrades gracefully rather than breaking: under stochastic replay each branch reward becomes a distribution, $C(d)$ a difference of means, and the measured null-replay distribution replaces the exact floor. The per-point recipe is: (i) estimate the null spread from n_0 null replays (the analogue of §4.1, now a variance rather than a maximum); (ii) size the per-branch replay count so that the minimal credit of interest exceeds the resulting standard error; (iii) cluster inference by task, as here. Exact screening-off becomes a test of $C_{HM} - C_M$ against the null band rather than a literal equality, and the coalescence lemma weakens to a coupling argument; what survives unchanged is the double-counting identity, which is algebraic and holds in expectation. The deterministic setting of this paper is the calibration case where all of this is measurable at zero variance — we deliberately paid throughput for that identifiability. We then executed this recipe (pre-registered, 180 replays): at temperature 0.8 with 12 seeds per arm, on the two highest-$|C_H|$ screened points and the three pressure breaks of the mt6 census. Declared outcome s1 held: at both screened points the stochastic $D = \bar{C}_{HM} - \bar{C}_M$ covers zero (at one of them exactly — every one of 36 replays returns the same reward); at 2/3 break points D excludes zero with the sign predicted by the deterministic taxonomy (api_router $D = +0.46$, CI $[0.38, 0.58]$; l_log_parser $D = -0.15$, CI $[-0.15, -0.13]$, against null spreads of 0.00–0.04), while the third (l_vending_machine, the weakest deterministic break at $|C_{HM} - C_M| = 0.09$) dissolves into the sampling noise. The deterministic census is thus not an artifact of greedy decoding: it predicts where the stochastic estimand separates from its null band.

F Training Acts 1–3 in Full

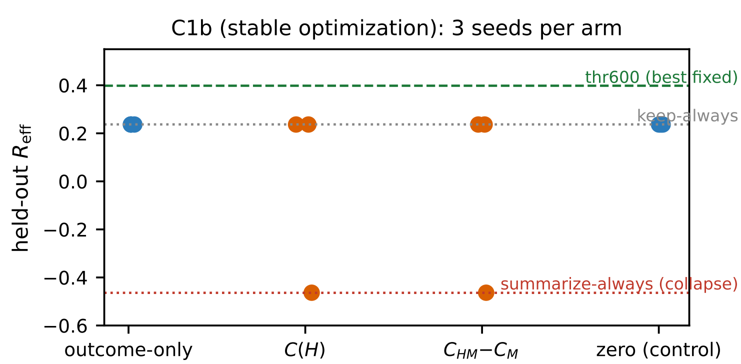

Figure 2: F5 — Training outcome (pre-reg 12, stable optimization). Held-out R_{eff}, 3 seeds per arm. No arm reaches the expressible thr600 policy; credit-driven arms (orange) additionally collapsed in 1/3 seeds each (anecdotal at $n = 3$).

Act 1: seed-dependent collapse. With raw features and lr $= 0.5$, every trained arm collapsed into one of the two fixed policies, with the attractor determined by seed (outcome: k/k/s; C_H: s/s/k; $C_{HM} - C_M$: k/s/s). GATE 3 failed. Two hypotheses remained: the credit estimator is broken, or the optimization is. A pre-registered audit recomputed 60 logged credits by fresh two-greedy-replay: 58/60 sign agreement (mean discrepancy -0.0015). Defining a harmful flip as logged-positive/recomputed-negative — a definition we state here, the registration specified sign agreement only — there are zero: the two disagreements are one benign flip (inventory_restock idx 0, $-0.084 \rightarrow +0.10$; logged as harmful, actually helpful) and one zero-to-negligible boundary case (c_temp_label idx 6, $0.0 \rightarrow -0.001$). The estimator is exonerated.

Act 2: stable optimization, no discovery. With the pre-registered fix (feature centering, lr = 0.1, gradient clipping), training is sane: gradients bounded, learned weights point the right way (negative bias, positive weight on context tokens, implicit decision thresholds at 2.5k–5.8k tokens). Yet no arm in any seed discovers the threshold policy: all non-collapsed runs converge to keep-always (held-out $R_{\text{eff}} = 0.237$ vs. thr600's 0.398; Figure 2). Credit-driven arms also retain collapse risk the outcome-only arm avoids (C_H: 1/3 seeds; $C_{HM} - C_M$: 1/3; outcome-only and zero: 0/3) — a seed-count difference that is anecdotal at $n = 3$ (Fisher $p = 1.0$) and that we report without testing.

Diagnosis. Dose-matching in total rollouts means credit arms pay for replays with episodes: across 3 seeds the outcome arm saw 1,789 training episodes, C_H 786, $C_{HM} - C_M$ 319 (zero: 1,910). One might suspect the credit arms simply saw too little data. The state-visitation and reward-margin records rule this reading out and localize the failure in three steps. First, the learned policies are behaviorally inert: their implicit decision thresholds sit at 2,500–5,800 context tokens, but no decision in any arm or seed — 0 of 13,498, including the zero arm, de facto keep-always, the policy that maximizes context growth (1,910 episodes, cap 1,532 tokens) — and no held-out decision (cap 1,200 tokens) ever reaches that region: the learned thresholds can never fire, so every non-collapsed policy is keep-always in behavior. Second, in the states the training tasks do visit (thr600 fires above 600 tokens, well inside the visited range), the reward margin between thr600 and keep-always is statistically indistinguishable from zero: paired bootstrap on the training tasks gives +0.024, CI95 $[-0.029, +0.079]$, $P(\text{diff} \leq 0) = 0.19$ (re-derivable from the raw calibration artifact by the bundled margem_calibracao.py, which fails on drift) — whatever gradient signal exists on the training tasks is bounded by this margin, which is undetectable at this sample size; we note the interval admits margins (up to +0.08) that more episodes might resolve, so this second argument is an underpowered null and the diagnosis rests primarily on the state-visitation argument. Third, thr600's held-out dominance comes from cumulative-cost structure: long held-out tasks accumulate prompt tokens under keep-always (e.g. l_vending_machine, 5,933 cumulative prompt tokens, $R_{\text{eff}} = -1.12$) — a cost regime the training distribution never makes consequential. The train/held-out split was alphabetical, fixed before training and before we knew this gap existed. The binding constraint is the training task distribution, not credit quality. This sharpens the design lesson: a cross-layer training benchmark must verify that its training tasks assign a nonzero reward margin to the states where candidate policies actually differ — otherwise credit fidelity is moot, and the failure will masquerade as an algorithmic one.

Act 3: verified margin, third attractor. We then ran the experiment this lesson demands (pre-registered with declared outcomes before any run). Stage A measured, for all 52 unique tasks in the project, the actual R_{eff} margin between thr600 and keep-always under two deterministic episodes per task, and selected the 16 tasks with positive margin (train/held-out split by alternating rank, fixed before training); held-out references: keep-always -1.181, thr600 -0.746, a verified margin of +0.435. Stage B re-ran all four arms × 3 seeds, dose-matched at 1,600 LLM calls, with the Act-2 optimization. The result is sharper than either prior act: every trained arm in every seed — outcome-only and both credit arms, 9/9 runs — converges to the same third attractor, summarize-always, scoring held-out $R_{\text{eff}} = -0.143$ with $R = 0.000$, while the zero arm (de facto keep-always) solves 45% of held-out tasks at $R_{\text{eff}} = -1.181$. The trained arms are right: on this pool of long, hard tasks under $\lambda = 25$, universal cheap failure is the true optimum of the declared objective — solving a task costs roughly its reward in tokens. Training worked; the objective was mis-specified. The declared outcome bin is s2 (all arms escape keep-always, no credit-vs-outcome ordering), and we own an omission: summarize-always is a fixed policy whose R_{eff} was computable from the stage-A machinery before stage B ran — the dominance of the third attractor was checkable analytically and we did not check it. The design lesson finalizes into its strongest form: verifying a margin between two named policies is not enough — the margin must be verified against every fixed attractor expressible by the policy class (here summarize-always dominates both named references), and the cost weight λ must leave solving profitable, or every signal — outcome or credit, at any fidelity — will honestly optimize its way to failure. A secondary observation at matched call dose: the credit arms reached the same endpoint from roughly 9× fewer episodes (outcome 756–825 episodes per seed; $C_{HM} - C_M$ 82–87), consistent with credit buying per-episode signal at the price of replays, but on this landscape there was only one place to go.

G Additional Figures and Tables

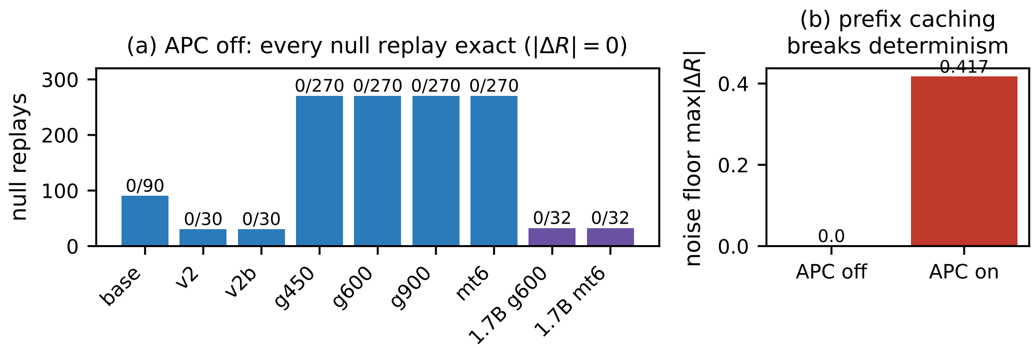

Figure 3: F1 — Measured noise floor. (a) Dedicated null-replay runs (Qwen3-4B blue, Qwen3-1.7B purple); together with nulls attached to interaction runs, the Qwen3-8B and MBPP+ replications, and the curated-pool runs, all 9,296 null replays are exact, floor = 0.0. (b) With vLLM automatic prefix caching on, the floor rises to 0.417 (7/270 inexact): greedy determinism breaks silently.

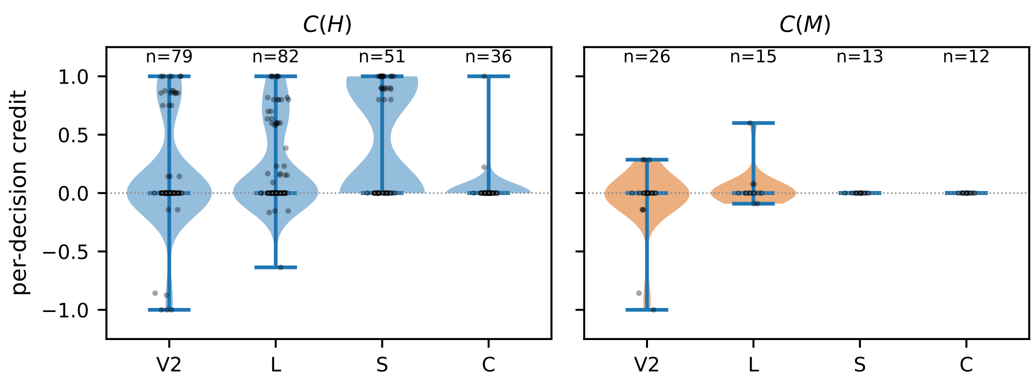

Figure 4: F2 — Per-decision credit by layer and stratum. C_H (248 points) and C_M (66 points) against an exact zero floor. Strata: V2/L information-destruction, S designed synergy, C controls.

H Critic Detail

Per-stratum detail for §5: on the combined L+V2 strata ($n = 161$, task-clustered Spearman) the GBM reaches 0.667 and the linear critic 0.675, against 0.708 for the position heuristic and 0.770 for |context|. Per stratum, L is a tie (GBM 0.747 vs. position 0.741), in V2 the heuristic wins (0.693 vs. 0.796), and S saturates for everything (AUROC 1.0). For detection ($|C| > 0$) the GBM has a modest edge (AUROC 0.846 vs. 0.735 for position and 0.785 for

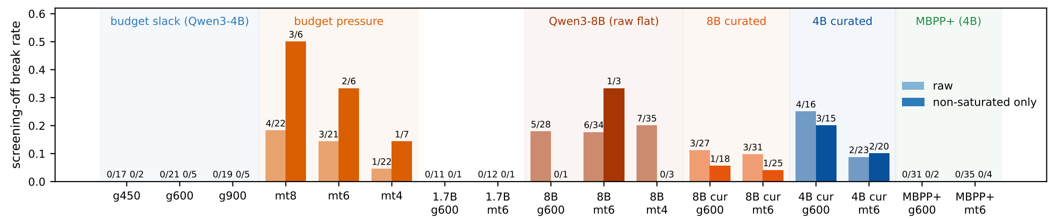

Figure 5: F3 — The slack/pressure contrast is itself a property of the triple: regime-dependent on the primary triple, flat or absent on others. Break rate of $C_{HM} = C_M$ per configuration, raw and conditioned on non-saturated points. Slack: 0/57 raw (0/12 non-saturated). Pressure: 8/65 raw (6/19 non-saturated, $p = 0.037$). Qwen3-1.7B (purple) replicates slack; its pressure null is explained by saturation (only 1 non-saturated point). Qwen3-8B (red) breaks raw screening at a flat rate across regimes (5/28, 6/34, 7/35), and the flatness survives the pre-registered curated pool (orange: 3/27 slack, 3/31 pressure); the symmetric control (dark blue) shows the 4B also breaks on the curated pool (4/16, 2/23) — the triple dependence of §8. MBPP+ (green, Qwen3-4B) replicates screening unconditionally (0/66).

the sign-inverted context-size heuristic) with overlapping CIs. The transfer test evaluated on the 94 MBPP+ C_M points (the only quantity with ground truth on both sides; C_H has none in the MBPP+ runs): Spearman 0.03 linear / −0.08 GBM, AUROC 0.30/0.50, MAE worse than a constant predictor.

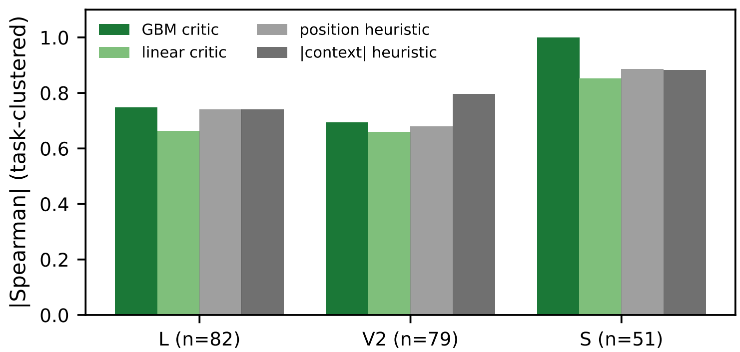

Figure 6: F4 — Critics vs. trivial heuristics per stratum (task-clustered |Spearman|). The critic never beats the best heuristic; C_H is structured by position and context size.

I Pre-Registration Ledger

All confirmatory analyses were pre-registered in the project's execution plan before the corresponding data were collected, with one dated exception. The ledger is an internal, same-team pre-specification, timestamped by git commits; for every item but one the registration commit precedes the corresponding run artifacts. The exception is item 10: the first slack census (g600) completed 34 minutes before the registration commit, so we treat g600 as discovery data and report the discovery/confirmation split in §4.3 (confirmation alone: 39/42). It is not a third-party registration, and minutes-to-hours gaps in a single-team pipeline cannot exclude unreported exploration before each commit (the pilot runs that preceded the formal ledger are preserved in the artifact tree). Its evidential force rests on the recorded failures, not on the registration mechanism itself. Table 7 lists all 34 pre-registrations with their outcomes, including the four that failed and the eleven partial. Failed predictions are reported in the main text where they occur.

J Null-Replay Reconciliation

Table 8 reconciles the canonical null-replay total (9,296, all exact) against every run artifact; a bundled script re-derives the table and fails if any count drifts. Excluded runs are listed with their reasons: the ASLR pre-fix runs (premises held, fidelity broken by an unsanitized memory address; root-caused and re-run) and the APC-contaminated run (premises violated by design, kept as the Figure 3b demonstration).

The ASLR incident in full. In a discarded pre-fix run of the first hardened pool (excluded from the 9,296: its pool failed the pre-registered curation rule, and its configurations were re-run after the fix), 2 of 432 nulls were inexact — the first fidelity breaks observed in this project (2/9,728 including every null ever run under the stated premises). Root cause, identified from the diverging trace: a memory address inside a Python repr in the pytest output (ASLR) entered the model's live context, and greedy decoding diverged downstream. The sanitizer now normalizes addresses at the source (regression-tested), and the affected configurations were re-run from scratch. Two honesty notes: the 5,023 nulls collected before the fix were exact because their pytest outputs happened to contain no address-bearing reprs — measured, not guaranteed by sanitization; and "four curated-pool configurations" means 2 budget regimes × 2 pool halves. The lesson generalizes: replay-grade determinism requires auditing every byte that reaches the context, not just the serving stack.

The concurrency incident. A scheduling error in our own pipeline briefly ran two experiment chains against the same server, and the very first affected cell produced a reproducible non-exact null ($\Delta R = +0.67$ twice, diverging at the second retry call). Concurrent batching, like APC, is sufficient to destroy the floor; the contaminated runs were quarantined and re-run sequentially, and the incident is preserved in the run ledger.

The LLM-summarizer cell. A pre-registered cell replaces the rule-based summarizer with a greedy LLM summarizer (the served model summarizes its own history under a fixed prompt, temperature 0), re-running the full pipeline on the same 30 tasks: all 317 null replays (270 dedicated + 47 attached to the census) reproduced R exactly. The premise "greedy summarization is deterministic" was declared falsifiable in the pre-registration; it held. The cell's census had low pivotal yield — 3 pivotal points out of 13 measured, 2 screened, 1 break (saturated) — because the semantic summary preserves the very constants whose destruction generates pivotal harness credit in this pool; we report the cell as underpowered for the screening question (its declared outcome bin) and informative for the floor question.

K Break-Point Anatomy (Descriptive)

This appendix reports the full descriptive trace anatomy of the curated-pool screening breaks (0 new rollouts; all from recorded replay traces). It is descriptive throughout: no count below is a pre-registered test.

The 8B's six breaks. One uniform base and two patterns on top of it. Uniformly: every break is a keep→summarize flip at a write_file turn, the summary destroys essentially all critical constants in context (8/8, 5/6, 7/7, 10/10), and the harness flip alone is destructive. On top: either the sampled a' carries some constants and partially rescues the destroyed context (the joint flip lands between the single flips: h_hotel_folio, h_cargo_manifest — the former tracks the model, $C_{HM} = 0.08$ vs. $C_M = 0.00$; the latter the harness, $C_{HM} = 0.46$ vs. $C_H = 0.62$), or a' is itself beneficial and the destruction annuls exactly that benefit (the joint flip returns to the original reward: x_hours_bank, h_sku_validator — the one positive I is of this kind). The two recurring points are numerically identical across regimes — same four arm rewards, same destroyed constants, same free-divergence offsets, from different baseline trajectories — so the break is a property of the decision point (task × turn), not of the budget regime.

The 4B's curated-pool breaks (pre-reg 19). The same anatomy adds a structural asymmetry: two of the 4B's four unique break points (three of the six break instances, counting the recurring h_turnstile_fsm point) are restore-direction flips (summarize→keep: the baseline had already summarized, and the flip restores destroyed constants), a direction absent from the 8B's breaks, and they follow a third pattern — individually beneficial or neutral flips whose joint application destroys the benefit ($I > 0$). Only one 4B point falls into an 8B pattern.

Audit details. The recurring h_hotel_folio point is numerically identical across regimes in all four arm rewards despite distinct baseline trajectories, and the recurring h_turnstile_fsm magnitude difference ($I = +0.77$ slack vs. $+0.31$ pressure) traces entirely to baseline quality — C_{HM} and all free-divergence offsets are identical across regimes. The I magnitude at a recurring point is therefore baseline-relative even when the break itself is stable.

L Design Brief for a Cross-Layer Credit Benchmark

The training negatives in the ledger are constructive: each names a property a benchmark must certify before a cross-layer credit comparison on it is interpretable. A pool qualifies only if it offers, per task:

1. Deterministic fractional reward (pre-regs 1, 27): a test suite scoring strictly inside $(0, 1)$ on the buggy state, deterministic at generation — otherwise the null floor is unmeasurable and no C is interpretable.
2. Controllable position of destroyable information (pre-regs 6, 16/20/25): the information a harness decision can destroy must be placeable relative to the decision point, with per-stratum calibrated recoverability; public benchmarks provide neither, which is why our external replications could test the floor and screening but not the regime contrast.
3. Verified attractor margins (pre-regs 32–34): for the declared policy class and cost weight, enough tasks (our gate: ≥ 10) on which the best conditional policy beats every expressible fixed attractor by a margin resolvable at the planned budget (≥ 0.10 in R_{eff}). Pre-reg 34 shows this must be verified, not assumed: on our V2 stack no cost weight yields even 3 such tasks, and training collapses regardless of learning rate or budget.
4. Cost weights scaled to the stack (pre-reg 32): λ grids do not transport across token-magnitude regimes; the benchmark must publish per-stack calibration data (fixed-attractor sweeps), not a single constant.
5. A screening census with registered accounting (pre-regs 29, 30): pivotal-point discovery, per-type screening rates, and the gate of Corollary 3 — with denominator and treatment of degenerate last-mover points fixed in advance, since the verdict can be sensitive to both (Table 3).

Items 1–2 make credit measurable; items 3–4 make training on it adjudicable; item 5 decides whether cross-layer correction is worth buying at all. Our designed pools satisfy 1–2 and 5 but — as pre-reg 34 established post hoc — not 3; to our knowledge no existing benchmark satisfies all five.

Item 3 is harder than it looks: after pre-reg 34 we attempted to construct a passing pool adversarially, with full knowledge of the mechanism (24 tasks: information poisoned against summarization paired with context-heavy tasks demanding it). Four design iterations, each calibrated against the real model, all failed the gate — and each failed through a distinct behavioral pathology rather than a task parameter: the model reads too little for the poison to bite the default policy; multi-stage repairs exceed its competence band entirely (all policies sink together); two-source synthesis collapses the pool; and in the final iteration, summarization mid-repair induced a degenerate loop (the model rewrites the same file without ever re-running tests, overflowing the context under every context policy). The margin of item 3 must be verified against model behavior, not estimated from task structure — exploratory, single-seed; the record is in the released ledger artifacts.

A follow-up on the final pool with a 2× larger model (Qwen3-8B) sharpens the requirement rather than relaxing it. The looping pathology disappears, four tasks are solved by exactly the designed mechanism (keep overflows, summarize repairs, margin +0.95 each), and the conditional target strictly dominates every fixed policy in mean efficiency (+0.16) — the first landscape where it does. Yet the registered gate still vetoes (0/24): the conditional oracle commits at its first consulted decision and thereafter reproduces one fixed policy token-exactly per task, so its per-task margin over the best fixed policy is identically zero. A target that exercises only across-task conditionality can never pass a per-task-versus-best-fixed criterion; item 3's margin must therefore be specified per attractor (against each fixed policy the optimizer can collapse into) and at the level the training signal observes — mean efficient reward — with any such revision registered before training, not after.

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Claim	Evidence	Status
Replay noise floor is exactly zero	9,296/9,296 nulls exact, 3 models × 3 envs (Table 8)	confirmed (pre-reg 1)
Floor survives an LLM-in-the-loop summarizer	317/317 nulls exact (greedy summarizer cell)	confirmed (pre-reg 22)
Deterministic taxonomy predicts stochastic estimand	temp 0.8: 2/2 screened cover 0; 2/3 breaks exclude 0, predicted sign	confirmed (pre-reg 23, outcome s1)
Training negative survives a verified margin	9/9 trained runs find the true optimum of the mis-specified objective (cheap failure)	confirmed (pre-reg 24, outcome s2)
Under a sane objective, replay credit loses to outcome at matched dose	Act 4: arms separate 3/3 seeds; corrected credit removes the gradient its bias provided; episode-matched control: outcome $\geq C_H$ 3/3 seeds at equal episodes	identified (pre-reg 27, 31)
Zero floor survives an agent-class change	V2 mini-SWE (multi-file, 5 decision types, plain-text protocol): 117/117 nulls exact	confirmed (pre-reg 29)
All five harness decision types are reward-pivotal	V2 screening: 114 pivotal points across all 5 types (incl. retry 2/2, test-schedule ΔR to -0.64)	confirmed (pre-reg 29)
The census gate has both branches	V1 stack: closed, confirmed by 4 training acts; V2 stack: outcome s3 (no type screens $\geq$0.90); gate opens under 3/4 accountings (0.50 measured w/ duals to 0.13 pivotal w/o duals, §7)	measured; accounting-sensitive (pre-reg 29)
V2 licensed training collapses onto a fixed attractor	8/9 learning runs converge token-exactly to summarize-always (held-out 0.831 vs. 0.889 dominant target), one outcome seed to keep-always (-0.092), before credit ordering is testable; first run aborted at calibration (λ does not transport across stacks)	identified (pre-regs 32–33, outcome s3)
Action envelope is a harness decision with estimand-level effects	JSON envelope: 0 writes in 437 calls; plain-text: sampler near-deterministic at temp 0.8 (87/114 fail), escalation registered	demonstrated (pre-reg 28a, 29b)
vLLM APC breaks greedy determinism	floor 0.417, 7/270 inexact	demonstrated
Slack screening-off, 4B original pool	0/57 raw; pivotal 53/56 = 0.946, CI95 $[0.87, 1.00]$ task-clustered	internally consistent (pre-reg 10; endpoint re-specified post-hoc, §8)
Screening survives a' re-sampling	99/101 informative slack draws (CI95 $[0.94, 1.00]$); mt6 breaks persist 9/9	confirmed (pre-reg 21)
Screening completeness depends on the estimand	a'_s (summarized state): 24/31 slack, CI95 $[0.69, 0.88]$; de-screened points match the re-injection signature	measured; claim scoped (pre-reg 26)
V2 census verdict survives its estimand controls	re-draws: point classification 17/23 stable but s3 + open gate hold under both schedules (0.474/0.368 vs. 0.395); under a'_s screening drops to 8/21 and all non-screened points turn exactly additive ($I = 0$, $C_H \neq 0$)	held; claim scoped (pre-reg 30, r2 + b3)
Screening survives discovery/confirmation split	discovery (g600) 14/14; confirmation 39/42; strictly post-registration 27/30	confirmed (split forced by ledger audit)
Regime contrast not explained by selection composition	retained 8B points shallower than 4B's (turn 0.65 vs. 1.55); direction disfavors the contrast	supported (§8)
$C_H = -I$ at screened points	algebra (Prop. 1); exact in all measured slack points	identity
Pressure breaks recur at fixed points (4B)	8/65; same points across configs	point-level; task-clustered weak ($p = 0.125$)
Break rate monotonic in pressure	inverted U, Cochran–Armitage $p = 0.38$	failed (pre-reg 13)
Saturation mediates the window	conditioned rates only	descriptive (post-treatment)
Screening-off is universal across models	1.7B: 0 raw breaks; 8B: breaks flat in both regimes	refuted
Shield is a property of the (model, harness, task) triple	curation + symmetric control; break sets point-level disjoint across models	supported; positive characterization inductive
Re-injection mechanism generalizes	23/24 annotated (original pool) vs. 0/15 exposed (curated), Fisher $p = 1.0/0.94$	failed (pre-reg 18)
Screening replicates in a second environment	MBPP+: 11/11 pivotal (4B), 24/24 pivotal (1.7B); rest screens by construction; pressure cell empty	confirmed; pressure inconclusive (pre-reg 16, 20)
Screening replicates in a third environment	HumanEval+: 42/42 pivotal screened, 0 breaks; floor 1,294/1,294 exact	confirmed; pressure inconclusive (pre-reg 25)
Interaction-corrected credit trains a better harness	no arm learns thr600; margin +0.024, CI95 $[-0.029, +0.079]$	failed (pre-reg 9, 12); cause identified
Learned critic beats cheap heuristics	context-size heuristic wins overall	refuted; $C_H \approx$ position + size
Amortized credit transfers across environments	Spearman 0.03/-0.08, chance	refuted (pre-reg 16)
V2 collapse is a landscape property, not an optimizer artifact	analytic gate: no λ gives $\geq$10 tasks with attractor margin $\geq$0.10 (max 3); probe at lr/5, budget×2 still collapses token-exactly, $\theta \approx$	held (pre-reg 34, both parts)

Table 4: T0 — master claim ledger: every claim, its evidence, and its epistemic status. Bold failed/refuted rows are reported as results, not omissions. A bundled make reproduce

Act	Registered question	Identified failure	What it fixed for the next act
1	credit > outcome? (GATE 3)	optimization: lr 0.5 collapses every arm to a seed-dependent fixed policy; audit (58/60) exonerates the estimator	normalized features, lower lr, entropy floor
2	same, repaired optimizer	distribution: training tasks never visit states where policies differ (stable learning, no separation)	margin-verified task selection
3	same, verified margin	objective: mis-specified λ — all 9 runs correctly find its true optimum, universal cheap failure	analytic λ^* verified against all fixed attractors
4	same, sane objective	none — arms separate 3/3 seeds: outcome $0.440 > C_H$ $0.405 >$ corrected $0.392 =$ untrained	the closed branch of Prop. 2, confirmed

Table 5: The four training acts. Each failure was diagnosed by a pre-registered audit before the next act ran.

Method	counterfactual via replay	measures C_H	measures C_M	I on same trajectory	trains with I	structure across models tested
CHILL-Harness	✓[off]	✓	–	–	–	–
CAR	✓	–	✓	–[s]	–	–
C3 v2	✓	(per agent)		–	–	–
Replay audit (Zhang, 2026)	✓	–	✓	–	–	∼[m]
Co-Harness / HASE	–	–	–	–	–*	–
Ours	✓	✓	✓	✓	tested†	✓

Table 6: T1 — positioning. [off]Checkpointed paired replay used offline to train an amortized estimator; disabled at decision time. [s]CAR measures step–step interaction (Shapley) within one layer, not across the model/harness boundary. *Joint harness+weights training without per-decision causal measurement. [m]Observes that counterfactual measurability is model-dependent, within a single layer; no cross-layer structure. †Tested with dose-matched controls and pre-registration; the outcome is an identified negative (§6).

#	Pre-registered prediction / rule	Outcome	Status
1	Noise floor per config = max$\lvert\Delta R\rvert$ over nulls	0.0 in all 26 dedicated + 24 attached runs (9,296 nulls; Table 8)	held
2	Phase-A yield / stopping rule	followed as written	held
3	Critic evaluation statistic (task-clustered Spearman)	mixed result reported as-is	held
4	Pre/post feature split for critic	followed	held
5	Structured s' = distinct estimand, reported separately	followed	held
6	C1 arms dose-matched in total rollouts	followed	held
7	Credit directions never aggregated	followed	held
8	Saturation analyses outside confirmatory set	followed	held
9	GATE 3: arm 3 must beat arms 2 and 4	not met	failed
10	Screening-off mechanism sign test	71/78 as registered; re-specified post hoc to pivotal points (non-pivotal screen by construction): 53/56; g600 census predates this registration (discovery) — confirmation subset 39/42	held
11	Estimator audit: sign agreement on 60 recomputed credits	58/60, 0 harmful flips	held
12	C1b primary outcome: held-out $R_{eff} > 0.35$, $\geq 2/3$ seeds	no arm reached it	failed
13	Dose-response: monotonic break rate in pressure	inverted U ($p = 0.38$)	failed
14	D2c replication: floor, slack, pressure on Qwen3-1.7B	floor+slack replicate; pressure inconclusive	partial
15	D2d replication: floor, slack, pressure on Qwen3-8B (incl. mt4 escalation rule)	floor replicates; raw screening does not (5/28 slack breaks, shield inversion); 1 non-saturated pressure break	partial
16	D3 second environment: floor, screening, pressure on MBPP+ (60 tasks); critic transfer A→B	floor+screening replicate (66/66); pressure inconclusive (saturation); transfer null	partial
17	D4 curated-pool break rate on 8B: curation rule (baseline reward in (0.05, 0.95)), ≥ 12 of 24 hardened tasks	7/24 fractional (17/24 solved at 1.0); curation failed as pre-specified; run also exposed the ASLR fidelity leak (§4.1)	failed
18	D4b (final attempt): union pool ≥ 12 in window; (a) floor 0.0; (b) breaks ⇒ no re-injection (Fisher); (c) pressure rate $\geq$ slack	curation passed (22/44); floor held (792/792 exact); (b) failed ($p = 1.0$, $p = 0.94$); (c) failed (9.7% < 11.1%, flat); in-window breaks replicate incl. non-saturated slack	partial
19	D5 symmetric control: 4B on the 22 curated tasks: (a) floor 0.0; (b) 4B saturated-in-failure ($\geq 50\%$); (c) declared outcomes c1/c2/c3	floor held (396/396); (b) failed (1/16, 3/23 saturated); outcome c2: 4B also breaks in-window (4/16, 2/23) ⇒ claim re-scoped to (model, harness, task) triple	partial
20	D6 empty cell: 1.7B on MBPP+: (a) floor 0.0; (b) power gate (≥ 5 non-saturated); (c) declared outcomes c1/c2/c3	floor held (1,080/1,080); outcome c3: 2/22 non-saturated, pressure never binds — inconclusive as pre-specified; bonus: screening 44/44 raw in both regimes	partial
21	s' re-sampling sensitivity: 56 pivotal instances × 3 disjoint seed schedules; declared outcomes stable (≥ 0.90) / partial / fragile (< 0.75) on informative slack draws	stable: 99/101 (0.980, CI95 [0.94, 1.00]); 79/81 on genuinely new s'; null breaks persist 9/9	confirmed
22	LLM-summarizer cell: (a) floor 0.0 under a greedy LLM summarizer (declared falsifiable); (b) screening ≥ 0.90 / 0.75–0.90 / <0.75	(a) held (317/317 exact); (b) 2/3 pivotal screened, but yield too low to bin — semantic summaries preserve the constants that generate pivotal C_H in this pool	partial
23	Stochastic protocol at temp 0.8: 5 mt6 points × 12 seeds × 3 arms; s1 = screened cover 0 (2/2) and breaks exclude 0 ($\geq 2/3$)	s1: screened 2/2 cover 0; breaks 2/3 exclude 0 with predicted sign; weakest deterministic break (0.09) dissolves	confirmed
24	Margin-verified training rerun: stage A selects tasks with measured tie60(θ)-keep margin (14/52, held-out margin +0.435); stage B 4 arms × 3 seeds at 1,600 calls; outcomes s1 (credit > outcome) / s2 (escape, no ordering) / s3 (keep-always again)	s2: all 9 trained runs converge to a third attractor (summarize-always, $E_{eff} = -0.145$, $R = 0$) that truly dominates the declared objective; λ mis-specification exposed	partial
25	D7 third environment: HumanEval+ (60 tasks, generator frozen pre-run): (a) floor 0.0; (b) slack screening replicates; (c) pressure outcomes c1/c2/c3	floor held (1,284/1,284); slack and pressure: 42/42 pivotal screened, 0 breaks; (c) inconclusive by saturation as pre-specified (2 non-saturated)	partial
26	Estimand control: s_r' from the summarized state (re-injection impossible), all 44 keep-vs-summarize pivotal instances; outcomes s1 (≥ 0.90 screening) / s2 (0.75–0.90) / s3 (<0.75) on informative slack pairs	s2: 24/31 = 0.774, CI95 [0.69, 0.86]; all 7 de-screened points show $C_M = 0$, $C_{HM} \approx C_H$ (re-injection channel off); both in-population null breaks screen under s_r'	partial
27	Act 4 same objective: analytic λ^* from fixed grid + margin verified against all fixed attractors (12 tasks); outcomes s1 (credit > outcome) / s2 (learning, no credit ordering) / s3 (collapse); secondary: corrected arm $\geq C_H$ arm	s2, first positive: outcome 0.440 > C_H 0.405 > $C_{HM} - C_M$ = keep 0.392, 3/3 seeds; secondary failed inverted — the bias is load-bearing (§6)	partial
28	V2 mini-SWE pilot, 4 staged gates: pool ≥ 16 validated tasks; success in [0.30, 0.70] with <15% malformed; determinism (3/3 reruns byte-identical, 5/5 nulls exact); ≥ 3 of 5 harness decision types pivotal. Two recalibrations pre-registered before re-run (28a: plain-text action protocol after the JSON envelope yielded 0 write actions in 437 calls — itself a reportable harness effect; 28b: flip-candidate selection)	GO on all 4: pool 20/20; success 0.50, 0% malformed; determinism exact; 3 types pivotal (context 3/8, observation 4/5, termination 1/5; test-schedule 0/5 reported as non-pivotal)	held
29	V2 multi-decision census on a frozen 60-task pool, 2 configs (slack/pressure): Tests 0 V2 floor; per-type selective screening; $C_H/C_M/C_{HM}/I$ at pivotal points; declared outcomes s1 (screening is type-dependent) / s2 (screening dominates all types) / s3 (general break); pre-registered training gate: F2–F5 only if $\geq 20\%$ non-screened pivotal mass (Corollary 3 applied prospectively). Addenda 29a (terminal flips make the HM arm analytically degenerate) and 29b (s' sampler escalation to temp 1.2, stratified analysis, 0.15 divergence rule) registered mid-run before rerunning	floor 117/117 exact; all 5 types pivotal (114 points); outcome s3: no type screens ≥ 0.90 (context 0.609, observation 0.667, termination 0.100, test-schedule 0.333); gate opens under 3/4 accountings (0.50 measured to 0.13 pivotal-denominator w/o degenerate duals — the registration under-specified the accounting, reported in §7); strata diverge by 0.084 < 0.15 (pooled primary)	held (amended mid-run)
31	Episode-matched outcome-only control for Act 4 (the flagged confound): the deterministic outcome-only θ-trajectory sliced at the credit arms' episode counts (139/139/137 and 70/68/70), evaluated under the identical held-out protocol; validity gate: re-evaluating the final θ must reproduce 0.4402 exactly; declared outcomes s1 ($\text{outcome}_{ep} > \max C_H$ in $\geq 2/3$ seeds: reading survives) / s2 (below pre-reg C_H in $\geq 2/3$: sample-size artifact) / s3 (attractor-level)	gate passed (exact equality); s1: 0.416/0.250/0.398 vs C_H 0.405/0.405/0.398 — $\geq C_H$ 3/3 seeds, strictly above 2/3, seed 3 ties exactly; at 70/68/70 episodes outcome still beats the corrected arm's attractor 2/3	confirmed
32	V2 training (open branch): accounting fixed ex ante (measured points w/o degenerate duals, 15/38 = 0.39, gate opens); Act-4 protocol — calibrate 3 fixed attractors on all 60 tasks, select λ^* and a margin-verified pool analytically before any training (abort if < 10 eligible); 4 arms × 3 seeds, dose- and episode-matched; context overflow = failed episode with paid tokens (amendment 32a, registered before rerunning)	aborted at the registered criterion: ≤ 3 eligible tasks under $\lambda \in [1, 25]$ — the grid transported from V1 is mis-scaled to V2 token magnitudes; no training data touched	held (abort rule fired)
33	V2 training, re-registered: single change — λ grid rescaled to [0.02, 0.5], ties break to larger λ; all else inherited from 32; declared outcomes s1 (credit > outcome under both accountings) / s2 (learning without credit ordering) / s3 (no arm beats the best fixed attractor)	$\lambda^* = 0.2$, 10 tasks; s3: 8/9 learning runs collapse token-exactly onto summarize-always (0.831 vs. 0.889 target); one outcome seed and the untrained control at keep-always (-0.092); episode-matched slices leave the bin unchanged (one transient mixed-policy slice at 0.361)	held
30	V2 estimand controls mirroring V1 items 21/26 (number reserved earlier; registered and run after 31–33): (A) s' re-sampling on all 48 measured census points, two disjoint seed schedules at each point's census temperature, point-classification stability + recomputed headline per schedule; (B) s_r' from the summarized state on keep-vs-summarize context points; declared outcomes s1/s2/s3, b1/b2/b3	s2: classification draw-sensitive (17/21 stable) but s3 verdict and open gate survive both schedules (0.474/0.368 vs 0.395); b3: screening 8/21 = 0.381 under s_r', all 13 non-screened points exactly additive ($I = 0$, $C_M \neq 0$)	held; claim scoped
34	Post-mortem of 33's collapse, two parts. (A) analytic power gate: license training only if ≥ 10 pool tasks have attractor margin ≥ 0.10 in $R_{eff}(\lambda)$ at some λ (margins from 32's calibration data; design motivated by exploratory inspection, declared as such); (B) optimizer probe: outcome + corrected arms, 1 seed, lr 5× smaller, budget 2× larger, same pool/λ^*; declared outcomes p1 (both collapse) / p2 (arms separate) / p3 (one collapses)	(A) no λ yields ≥ 10 eligible tasks (max 3 at margin 0.10; 8 at 0.08) — the gate would have vetoed 33; (B) p1: both arms collapse token-exactly onto opposite attractors with $\theta \approx 0$ ($\lVert\theta\rVert_\infty \leq 0.035$) — landscape, not optimizer	held

Table 7: Pre-registration ledger. Failures are results, not omissions: #9 and #12 constitute the two-stage training negative (§6); #13 led to the competence-window analysis (§4.3).

Run	Model / pool	Dedicated	Attached	Inexact
g450, g600, g900	4B / original, slack	3 × 270	64, 61, 61	0
mt8, mt6, mt4	4B / original, pressure	3 × 270	61, 61, 61	0
v2, v2b	4B / pilot pools	2 × 30	16, 16	0
q17_g600, q17_mt6	1.7B / original	2 × 270	32, 32	0
q8_g600, q8_mt6, q8_mt4	8B / original	3 × 270	78, 78, 78	0
v4_g600, v4_mt6	8B / hardened v4 (post-fix)	2 × 216	—	0
v5_g600, v5_mt6	8B / v5 pool (post-fix)	2 × 180	—	0
v5cur_g600, v5cur_mt6	8B / curated union	—	66, 66	0
q4cur_g600, q4cur_mt6	4B / curated (symmetric control)	2 × 198	53, 53	0
mbpp_g600, mbpp_mt6	4B / MBPP+	2 × 540	107, 107	0
mbpp17_g600, mbpp17_mt6	1.7B / MBPP+	2 × 540	78, 78	0
he_g600, he_mt6	4B / HumanEval+	2 × 540	107, 107	0
ls600	4B / original, LLM summarizer	270	47	0
Canonical total		7,728	1,568	0
Excluded from the canonical total:				
v4_g600, v4_mt6 (pre-fix)	8B / hardened v4, ASLR leak	2 × 216	—	2
g600 (APC)	4B / original, APC on	270	68	7 + 2

Table 8: Run-by-run null-replay reconciliation. Canonical total $7{,}728 + 1{,}568 = 9{,}296$, all exact; including the pre-fix runs, 2/9,728 nulls ever run under the stated premises were inexact (both in one pre-fix run, §4.1). "Attached" nulls are the queue-floor checks run inside each interaction experiment.